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MASTER THESIS

Interpretable Machine Learning for Urban Housing Affordability: A Data-Driven Analysis of U.S. Metropolitan Markets

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Abstract

Housing affordability has shifted from a temporary fluctuation to a persistent structural challenge across U.S. metropolitan areas, yet affordability, rental pressure, and market competitiveness are often examined in isolation, and machine learning applications in this domain tend to prioritise predictive accuracy over interpretability. This thesis develops an interpretable analytical framework that integrates these dimensions to explain housing price variation. A PRISMA-based review of the sustainability literature situates the study within the interdependencies of the Sustainable Development Goals, particularly SDG 11 (Sustainable Cities and Communities), providing a conceptual foundation that is then narrowed to a concrete, measurable urban domain. Building on this foundation, Zillow housing indicators covering 2018–2025 are analysed through exploratory data analysis, feature engineering, and predictive modelling. Linear Regression, Decision Tree, Random Forest, and Gradient Boosting models are developed and compared under a unified pipeline with consistent train–validation–test partitions to assess their ability to capture nonlinear relationships in housing prices. The tree-based ensembles consistently outperform the linear baseline across all evaluation metrics, and the tuned Gradient Boosting model achieves the strongest predictive performance, reaching an R^2 of 0.987 and a test RMSE of approximately \$19.5K. Across all models, affordability indicators, particularly YearsToSave and MedianRent, emerge as the dominant drivers of median sale prices, while short-term competitiveness measures play a comparatively minor role. Visual analytics and rule-based model interpretation provide transparent insights into housing market segmentation. Overall, the findings demonstrate that interpretable ensemble learning can deliver both accurate predictions and meaningful insight into housing price formation, offering a practical framework for analysing urban housing dynamics and supporting data-driven decision-making.

Keywords: Urban Housing Affordability, Tree-Based Machine Learning, Ensemble Learning, Interpretable Modelling, Data Analytics, Sustainability

1. Introduction

1.1. Background and Motivation

THE pursuit of sustainable development relies on the integration of economic, social, and environmental dimensions. Within this context, the United Nations Sustainable Development Goals (SDGs) provide a comprehensive and widely recognised framework for addressing global challenges such as inequality, resource management, and long-term societal well-being [38].

A key characteristic of the SDG framework is its interconnected structure. Rather than functioning as isolated objectives, the goals form an integrated system where progress in one area can influence outcomes in others [14]. However, despite this systemic design, existing policy and analytical approaches often fail to adequately capture these interdependencies, leading to fragmented decision-making and missed opportunities to leverage synergies or mitigate trade-offs [26, 5].

Despite advances in data analytics, translating sustainability frameworks into measurable and actionable insights remains challenging. The assessment of Sustainable Development Goals (SDGs) relies heavily on the availability and quality of data. However, data gaps and measurement limitations continue to constrain effective sustainability assessment and decision-making [25]. Such limitations make it important to focus on domains where sustainability challenges can be examined more directly.

Housing plays a central role in sustainable development, particularly within SDG 11 (Sustainable Cities and Communities), as it directly affects affordability, social equity, and urban resilience [37]. As such, it provides a meaningful setting for examining how sustainability challenges emerge at the urban level.

At the same time, rapid urbanisation and rising housing costs have intensified affordability challenges in metropolitan areas worldwide, transforming housing into one of the most pressing global urban concerns of recent years [39]. These dynamics highlight the need for analytical approaches that can capture complex relationships and support a better understanding of housing market behaviour.

1.2. Scope and Problem Statement

While global sustainability frameworks provide valuable conceptual insights, their broad scope often limits their applicability in data-driven and decision-oriented analysis. This research addresses this limitation by narrowing its focus to urban housing markets in the United States, a domain where economic conditions, social equity, and spatial dynamics can be examined through measurable indicators.

Urban housing markets are characterised by rising property prices, increasing rental pressure, and significant regional disparities, with U.S. metropolitan areas in particular exhibiting strong heterogeneity in affordability conditions driven by differences in housing supply and local economic factors [12]. These dynamics create affordability challenges, shaping both market behaviour and residential decision-making across metropolitan regions.

Traditional linear models often struggle to capture nonlinear relationships and complex interactions between variables in housing datasets, limiting their ability to represent real-world housing dynamics effectively [2]. Tree-based methods provide a suitable balance between predictive performance and interpretability in this context.

1.3. Research Aims and Objectives

Building on the sustainability context and problem definition outlined above, this section presents the research goal, aim, and objectives of the study.

Research Goal The long-term goal of this research is to contribute to a better understanding of urban housing affordability dynamics and to support data-driven decision-making in the context of sustainable city development.

Research Aim This thesis aims to develop an interpretable analytical framework for analysing housing price dynamics, with a particular focus on affordability-related factors, by integrating visual analytics and tree-based machine learning methods.

Research Objectives To achieve this aim, the following specific and measurable objectives are pursued:

- Identify and quantify the key economic and market indicators influencing housing prices, including affordability, rental pressure, and competitiveness metrics.
- Develop and evaluate interpretable tree-based machine learning models, including decision trees, random forests, and gradient boosting, to capture nonlinear relationships and interaction effects in housing data.
- Compare model performance using various quantitative evaluation metrics to assess predictive accuracy and generalisation.
- Apply visual analytics techniques to support model interpretation and identify meaningful patterns in housing market data.
- Analyse how affordability-related variables contribute to price segmentation and market structure.

Quantitative Measures Model performance is evaluated using statistical metrics, providing an objective basis for comparing models and assessing predictive accuracy.

Economic and Societal Relevance From an economic perspective, the analysis contributes to understanding the factors influencing housing price formation across metropolitan regions. From a societal perspective, the findings provide insights into affordability challenges in urban housing markets.

1.4. Research Questions and Expected Outcomes

The following research questions guide the analysis:

- RQ1. What are the primary drivers of housing price variation across U.S. metropolitan regions, and how do affordability and market competitiveness indicators influence housing sale prices?
- RQ2. To what extent can tree-based machine learning models outperform traditional linear approaches in predicting urban housing prices?
- RQ3. How do housing prices and affordability metrics evolve temporally across U.S. regions between 2018 and 2025?
- RQ4. What interpretable and meaningful patterns can be extracted from tree-based models through visual analytics and rule-based interpretation?

Expected Outcomes The analysis is expected to evaluate the performance of tree-based models in capturing housing price dynamics in comparison to linear regression. Visual analytics and model interpretation techniques are expected to reveal the key drivers of housing prices and price segmentation patterns across U.S. metropolitan regions. Finally, the study is expected to identify temporal trends in housing prices and affordability between 2018 and 2025.

1.5. Thesis Structure

This thesis is organised into five chapters, each representing a distinct stage in a progressive, data-driven research pipeline that moves from a global sustainability context toward applied urban housing analysis.

State of the Art reviews the literature from three complementary perspectives: sustainability and SDG interdependencies, urban housing and affordability, and machine learning approaches in housing analysis. It examines how these domains intersect within the context of urban housing markets and identifies the research gap that motivates the study.

Working Methodology transitions from the global sustainability perspective to the applied domain of urban housing. It presents the Zillow datasets covering the U.S. housing market from 2018 to 2025 and describes the data preprocessing and cleaning procedures. The chapter then conducts exploratory data analysis through statistical, correlation, and temporal analyses. It also introduces the feature engineering process and the modelling approach adopted in the study, including the theoretical foundations of linear regression and tree-based machine learning methods.

Data Modelling and Results presents the development, evaluation, and comparison of the predictive models. A linear regression model is first established as an interpretable baseline, followed by decision tree, random forest, and gradient boosting models. Model performance is assessed and the results are interpreted in the context of housing affordability and market dynamics.

Discussion and Further Research synthesises the key findings of the study, addresses the four research questions, and outlines directions for future research.

1.6. Software Platforms

All practical work in this thesis was conducted using the SAS ViyaTM platform and Python. The SAS ViyaTM platform served as the primary analytical environment. The analytical workflow was implemented using several components, including SAS Data ExplorerTM for data exploration and dataset inspection, SAS Visual AnalyticsTM for visualisation, exploratory analysis, and feature engineering, and SAS Model StudioTM for machine learning model development, tuning, and evaluation. Python was used for data integration, pre-processing, and cleaning, as well as for parts of the exploratory data analysis.

2. State of the Art

This chapter draws on three complementary areas of literature: sustainability, urban housing and affordability, and machine learning approaches in housing analysis. Because these literatures differ in both scope and analytical focus, different review approaches are adopted. A systematic review following the PRISMA framework [23] is applied to the sustainability literature, which provides the conceptual foundation for the study. The urban housing and machine learning literature is examined through a focused review of studies most relevant to the analytical objectives of the research.

2.1. Literature Review Design

2.1.1. Sustainability Literature Review (PRISMA)

The literature was reviewed using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework [23], a widely used methodology for conducting transparent and reproducible literature reviews. The process was carried out in two phases: article collection and article selection.

Search Strategy The primary database used was Web of Science due to its focus on peer-reviewed scientific literature. Search queries combined terminology related to the Sustainable Development Goals, their interactions, and the use of data-driven methods, including:

- “SDG interactions”
- “SDG trade-offs and synergies”
- “Data-driven approaches to SDG measurement”

Filtering and Data Source Validation The following criteria were applied to ensure the quality and relevance of the selected literature:

- **Language:** Only English-language publications were retained.
- **Time frame:** The review focused primarily on studies published from 2016 onwards. A small number of earlier references were retained where relevant.
- **Source types:** The majority of selected sources were peer-reviewed journal articles. Books, reports, and conference papers were included selectively when they provided relevant insights.

- **Study relevance:** Selected studies were evaluated for their relevance to the review topic.

Article Selection and PRISMA Stages

The four standard PRISMA stages were followed and are summarised below:

- **Identification:** A total of 247 records were retrieved from Web of Science through the keyword searches described above.
- **Screening:** After removing 1 duplicate, the remaining 246 records were screened on the basis of title and abstract, and 130 studies unrelated to the research topic were excluded.
- **Eligibility:** Full texts were retrieved for the 116 remaining records, of which 7 could not be retrieved due to institutional restrictions or paywalls. The remaining 109 articles were assessed for methodological quality, data availability, and relevance. At this stage, 89 further studies were excluded: 78 were excluded after full-text review as insufficiently relevant to the research objectives, 6 due to methodological weaknesses such as insufficient data or unclear methods, and 5 as non-English publications.
- **Inclusion:** A final set of 20 studies met all inclusion criteria and formed the foundation for the sustainability review presented in the following sections. A final validation check was conducted to ensure consistency in paper selection and to reduce the risk of reviewer bias.

The complete article filtering and selection workflow is summarised in the PRISMA flow diagram presented in Figure 2.1.

2.1.2. Urban Housing and Machine Learning Literature Review

The literature related to urban housing and machine learning in housing analysis was examined through a focused review rather than a systematic review process. The objective was not to cover all available literature, but to identify and analyse the most relevant studies for the research aims.

The primary database for this review was Google Scholar, chosen for its broad coverage of academic literature across journals, conference proceedings, and reports. Search queries were organised around three thematic blocks:

- **Urban housing affordability and regional heterogeneity**, using search terms such as "housing affordability", "U.S. metropolitan housing markets", "regional heterogeneity", and "housing market dynamics".
- **Machine learning and data science in housing analysis**, using search terms such as "machine learning housing markets", "random forest housing prices", "gradient boosting housing prices", "real estate data analytics".
- **Interpretability in applied machine learning**, using search terms such as "interpretable machine learning" and "feature importance".

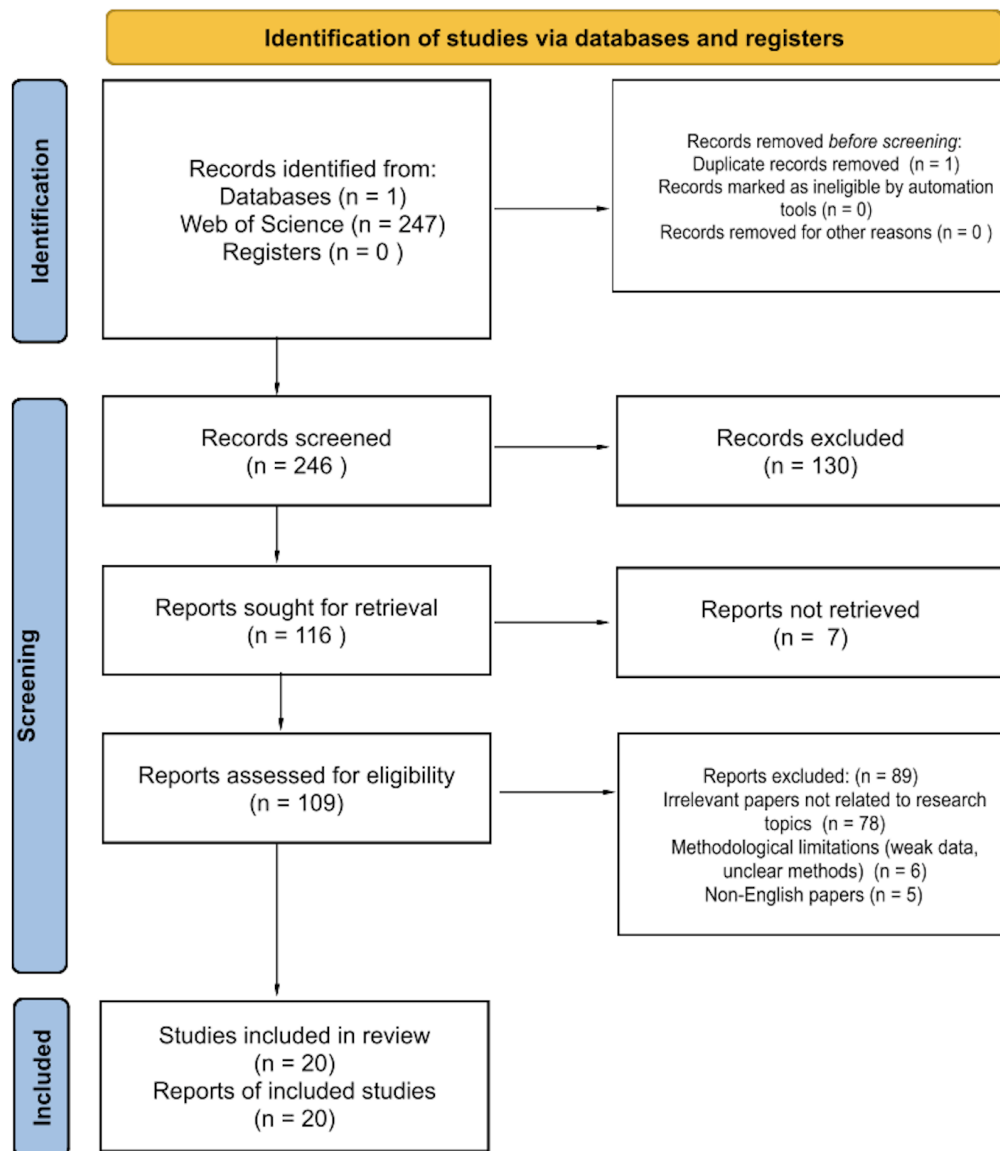


Figure 2.1.: PRISMA flow diagram for the study selection process of the sustainability literature review

Inclusion criteria prioritised peer-reviewed journal articles, with particular attention given to studies published from 2017 onwards. Preference was given to studies with empirical findings or methodological contributions directly relevant to U.S. housing markets and tree-based machine learning. Also, where possible, open-access versions were preferred. The studies identified through this process form the basis of the literature discussed in Sections 2.3 and 2.4.

2.2. Literature Review Analysis on Sustainability

2.2.1. The SDGs as an Integrated Framework

The commitment to achieving sustainable development involves balancing its economic, social, and environmental dimensions in an integrated manner [38]. Within this framework, SDG 11 (Sustainable Cities and Communities) is particularly relevant, as it directly addresses urban housing, affordability, and inclusive urban development.

In March 2015, the Open Working Group established by the UN General Assembly proposed 17 Sustainable Development Goals and 169 targets, along with a preliminary set of 330 indicators. Some goals build on the earlier Millennium Development Goals, while others introduce new concepts. The primary aim of the SDGs is to achieve a more sustainable and inclusive future by 2030 through coordinated action on poverty, inequality, human well-being, and environmental protection [38]. Progress is monitored through voluntary national reviews and global reporting frameworks, with particular emphasis on partnerships under SDG 17 [36].

A defining feature of the SDG framework is its integrated and indivisible nature. The goals are inherently interconnected, and progress in one area often affects outcomes in others [14]. In practice, implementation often lacks a structured approach to managing these interdependencies [33]. As a result, positive and negative interactions between targets are frequently overlooked, leading to fragmented strategies and unintended consequences [26]. Existing policy and evaluation frameworks also struggle to capture these systemic relationships, limiting their ability to support coordinated decision-making [5].

Measuring progress within such an interconnected system requires indicators that reflect these interdependencies. As highlighted by [27], identifying synergies and trade-offs is essential for effective sustainability assessment. In this context, indicators related to housing, affordability, and urban development can be interpreted not only as economic variables but also as components of a broader sustainability framework.

2.2.2. Synergies and Trade-Offs Among SDGs

Following the systematic selection process described above [23, 31], this section reviews the literature on SDG interconnections, with particular attention to interactions relevant to urban systems and housing dynamics.

Synergies Across SDGs

Many SDGs reinforce one another through positive interactions. For example, progress in SDG 1 (No Poverty) and SDG 2 (Zero Hunger) contributes to improved outcomes in SDG 3 (Good Health and Well-being), which in turn supports economic productivity and growth under SDG 8 (Decent Work and Economic Growth) [4, 27].

Similarly, improvements in SDG 6 (Clean Water and Sanitation) and SDG 7 (Affordable and Clean Energy) enhance quality of life, economic activity, and urban development,

particularly when aligned with sustainable consumption practices under SDG 12 (Responsible Consumption and Production) [9].

In urban contexts, SDG 11 (Sustainable Cities and Communities) plays a central role by linking infrastructure development (SDG 9 – Industry, Innovation and Infrastructure), social inclusion (SDG 10 – Reduced Inequalities), and environmental sustainability. Well-planned urbanization can improve resource efficiency and economic opportunities, but it also introduces challenges such as rising housing costs and spatial inequality [37]. These interactions highlight the importance of analysing urban housing not only as an economic issue but also as part of a broader sustainability system.

Trade-Offs in SDG Implementation

Despite these synergies, significant trade-offs between goals remain. Economic growth (SDG 8 – Decent Work and Economic Growth) and industrialization (SDG 9 – Industry, Innovation and Infrastructure) can increase resource consumption and environmental pressure, potentially conflicting with sustainable production (SDG 12 – Responsible Consumption and Production) and climate action (SDG 13 – Climate Action) [9].

Similarly, the transition toward low-carbon systems (SDG 13 – Climate Action) may create short-term disruptions in labour markets, particularly in sectors dependent on fossil fuels [20]. In urban settings, comparable trade-offs emerge between housing supply expansion, land-use regulation, and environmental sustainability, directly influencing affordability and accessibility.

The Stock-Flow Diagram presented in Figure 2.2 illustrates the complex interdependencies between social, economic, and environmental SDGs [18]. It highlights both reinforcing relationships and potential conflicts, emphasising the need for integrated approaches to sustainability analysis.

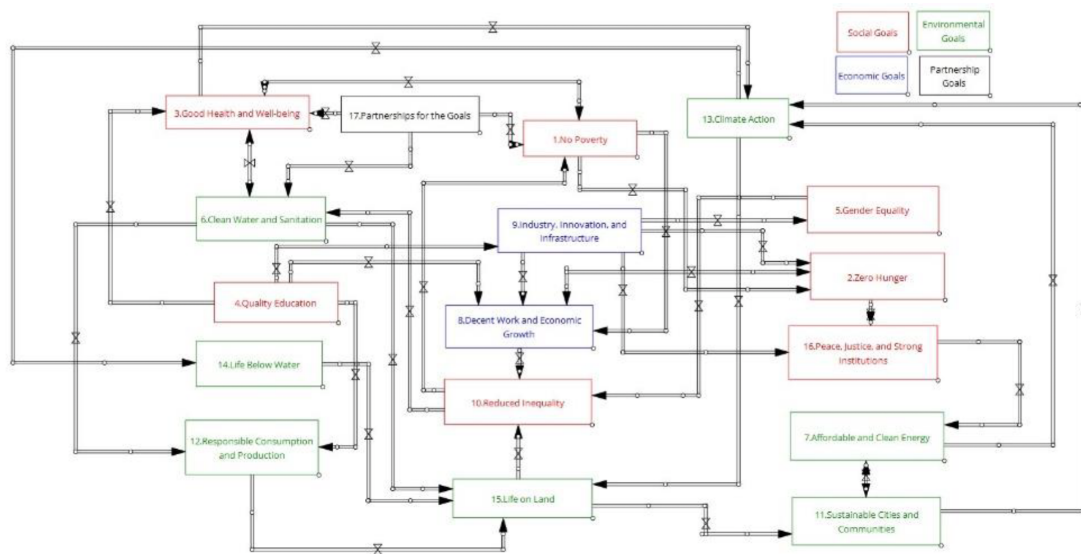


Figure 2.2.: Stock-flow diagram illustrating interdependencies among SDGs. Source: [18]

2.2.3. Implications for Data-Driven Sustainability Analysis

Effective monitoring of SDG progress depends on the availability of reliable, consistent, and comparable data, as well as appropriate analytical tools. Poor data quality, missing values, and limited accessibility remain major challenges, particularly when comparing results across regions and over time [25, 22].

The integration of data analytics plays a key role in addressing these challenges by enabling the extraction of meaningful patterns from complex and heterogeneous datasets. However, issues related to data governance, privacy, and security must also be considered when applying data-driven approaches [30].

Data availability remains uneven at the global level. Although the UN SDG database aggregates information from multiple sources, a significant proportion of indicators remains incomplete, especially in low-income regions [7]. This highlights the need for improved data collection, standardisation, and the use of alternative indicators where direct measurements are unavailable.

A wide range of methodological approaches has been used to analyse SDG interdependencies, including qualitative frameworks, system dynamics models, simulation techniques, and statistical methods, as summarised in Figure 2.3 [16]. Conceptual models help represent system-level relationships, while empirical methods provide measurable insights, although often with limitations in capturing causality. No single method is sufficient to fully represent these multidimensional interactions, which motivates the use of flexible analytical approaches.

Within this broader context, urban housing emerges as a key domain where economic, social, and spatial sustainability intersect. The following section focuses on the dynamics of urban housing markets and affordability, narrowing the scope from global sustainability frameworks to a concrete and measurable analytical setting.

2.3. Literature Review Analysis on Urban Housing and Affordability

Urban housing has become one of the most important topics in applied urban economics, mainly because affordability problems that were once seen as temporary have turned into long-term structural issues. In many U.S. metropolitan areas, housing prices and rents have increased faster than household incomes for more than a decade, shifting the understanding of affordability from a short-term fluctuation to a persistent challenge [39, 12]. This change is not only relevant to the research setting but also justifies the use of affordability indicators as key explanatory variables.

A major source of current affordability pressure comes from the imbalance between increasing demand and a slow supply response. In many economically productive regions, housing supply has not kept pace with demand, leading to a sustained increase in prices [12]. Housing supply responsiveness differs significantly across metropolitan areas due to geographic and regulatory constraints [29]. As a result, similar demand shocks can lead

Categories and assigned methods to analyse SDG entity interactions.

Category	Method	Refs.
Argumentative	Bayesian belief network (BBN)	(Hall et al., 2018)
	Causal loop diagram (CLD)	(Zhang et al., 2016)
	Cross-impact matrix (CI matrix)	(Allen et al., 2019a; Dawes, 2020; Kumar et al., 2018; Weitz et al., 2018; Zaini and Akhtar, 2019; Zelinka and Amadei, 2017)
	Structured elicitation of expert information (Expert)	(Allen et al., 2019a; Bhaduri et al., 2016; Hall et al., 2018; Hazarika and Jandl, 2019; Jaramillo et al., 2019; Waage et al., 2015; Wieser et al., 2019)
	Nilsson scale (N Scale)	(Allen et al., 2019a; Fader et al., 2018; Fuso Nerini et al., 2019; Hall et al., 2017; Hazarika and Jandl, 2019; Jaramillo et al., 2019; McCollum et al., 2018; Nilsson et al., 2016; Singh et al., 2018; Weitz et al., 2018; Zelinka and Amadei, 2017)
Literature	Non-systematic literature review (Non-syst)	(Alcamo, 2019; Bringeux, 2018; Fisher et al., 2017; Haines et al., 2017; Hazarika and Jandl, 2019; Manandhar et al., 2018; Morton et al., 2017; Pandey and Kumar, 2018; Recuero Virto, 2018; Swamy et al., 2018; Wydra et al., 2019)
	Semi-systematic literature review (Semi-syst)	(Bangert et al., 2017; De Paiva Serôa Da Motta, 2019; Engström et al., 2018; Fuso Nerini et al., 2019, 2018; Hanjra et al., 2016; Hepp et al., 2019; Schroeder et al., 2019)
Linguistic Simulation	Systematic literature review (Syst)	(Alcamo, 2019; Blicharska et al., 2019; Davide et al., 2019)
	Review of case studies (Case studies)	(Alcamo, 2019; Velis et al., 2017)
	Keyword analysis (KWA)	(De Paiva Serôa Da Motta, 2019; Le Blanc, 2015; Nugent et al., 2018)
	Agent based modelling (ABM)	(Wang et al., 2019)
	Computable general equilibrium models (CGE)	(Banerjee et al., 2019; Campagnolo and Davide, 2019; Doelman et al., 2019; Lucas et al., 2019; Matsumoto et al., 2019; Schütze et al., 2017)
Other quantitative	Energy system models (ESM)	(Engström et al., 2019; Vandyck et al., 2018)
	Integrated assessment models (IAM)	(Doelman et al., 2019; Fujimori et al., 2019; Gao and Bryan, 2017; Heck et al., 2018; Hutton et al., 2018; Lucas et al., 2019; Matsumoto et al., 2019; Obersteiner et al., 2016; Rao et al., 2016; von Stechow et al., 2016; Zhang et al., 2019)
	System dynamics modelling (SD)	(Allen et al., 2019b; Collste et al., 2017; Dawes, 2020; Pedercini et al., 2019, 2018; Spaier et al., 2017)
	Accounting framework (Account)	(Engström et al., 2018)
	Network analysis (NWA)	(Allen et al., 2019a; Dörgó et al., 2018; Feng et al., 2019; Jaramillo et al., 2019; Jiménez-Aceituno et al., 2020; Kuntić, 2019; Le Blanc, 2015; Lim et al., 2018; Lusseau and Mancini, 2019; Mainali et al., 2018; McGowan et al., 2019; Nugent et al., 2018; Sebestyén et al., 2019a, 2019b; Weitz et al., 2018; Zelinka and Amadei, 2017)
Statistical	Environmentally-extended multi-regional input-output models (IO)	(Hubacek et al., 2017; Scherer et al., 2018)
	Advanced sustainability analysis (ASA)	(Mainali et al., 2018)
	Autoregressive distributive lag bounds test (ARDL)	(Ngarava et al., 2019)
	Correlation analysis (Corr)	(Breach, 2019; Donaires et al., 2019; Kroll et al., 2019; Mainali et al., 2018; Ngarava et al., 2019; Pradhan et al., 2017; Sebestyén et al., 2019a, 2019b)
	Cox proportional hazards models (CPH)	(Akinyemi et al., 2018)
	Descriptive statistics (Descr)	(Howden-Chapman et al., 2020)
	Generalised method of moments (GMM)	(Matthew et al., 2019; Shahbaz et al., 2019)
	Joint correspondence analysis (JCA)	(Ulman et al., 2018)
	Linear mixed effect models (LMM)	(Lusseau and Mancini, 2019)
	Pairwise granger causality test (PGC)	(Ngarava et al., 2019)
	Principal component analysis and Factor analysis (PCA&FA)	(Donaires et al., 2019; Feng et al., 2019; Sen and Ongsakul, 2018; Spaier et al., 2017)
	Quantile regression, bootstrapped (Q Reg)	(Sinha et al., 2020)
	Regression analysis (Reg)	(Buonocore et al., 2019; Cluver et al., 2016; Hall et al., 2017; Malerba, 2019; Obersteiner et al., 2016; Ramos et al., 2018; Ulman et al., 2018)

Figure 2.3.: Overview of methodological approaches used in the literature to analyse SDG interdependencies. Source: [16]

to very different price outcomes across regions, which needs to be considered in empirical analysis.

These dynamics create strong spatial differences in housing outcomes. U.S. housing markets do not behave as a single unified system: high-demand coastal cities follow very different patterns compared to many Midwestern or Southern regions. High housing costs in productive regions limit household mobility toward higher-wage areas, slowing income convergence and increasing regional inequality [10]. At the same time, rental markets reinforce these patterns. As homeownership becomes less accessible, demand shifts toward rentals, increasing rents and extending the time required to save for a down payment [1]. From a data analysis perspective, this variation across regions is not noise but an essential part of the underlying structure.

At a more detailed level, housing prices are often modelled using the hedonic framework, which explains prices based on the contribution of individual property characteristics [21]. While this approach remains useful because of its interpretability, it relies on strong assumptions such as linear relationships and stable effects across different markets. In practice, these assumptions often do not hold. Housing prices frequently respond in non-linear ways, variables interact with each other, and different regions follow distinct patterns [17]. Affordability indicators are a clear example of this, as they often behave more like thresholds than smooth drivers of price changes. These limitations motivate the machine learning approaches discussed in the following section.

2.4. Literature Review Analysis on Machine Learning Approaches in Housing Analysis

The limitations of hedonic and other linear models discussed in the previous section have motivated a gradual shift toward more flexible, data-driven approaches in real estate and urban housing analysis. When prices depend non-linearly on affordability, when location interacts with structural attributes, and when market behaviour varies across metropolitan regions, models that assume a single additive functional form tend to misrepresent the underlying relationships. These features are not exceptional; they are common in modern housing data, which typically involves many variables, is heterogeneous across regions, and is characterised by strong interactions among them.

Machine learning enters this context as a direct response to these limitations. Mulainathan and Spiess [24] argue that ML methods are particularly suited to settings where the functional form is unknown and the goal is to approximate complex relationships between features and outcomes, rather than to recover a predefined parametric structure. Housing markets closely match this setting. Empirical studies comparing ML models with linear hedonic benchmarks report consistent improvements in out-of-sample accuracy, especially in heterogeneous urban submarkets [6, 15]. The approach has also been extended to applied tasks such as identifying underpriced properties from market listings [3].

Within this family of methods, tree-based models are particularly well-suited to housing data. Decision trees partition the feature space through a sequence of binary splits, which allows them to capture threshold behaviour and non-linear responses that frequently arise in housing markets. This is especially useful for affordability indicators

and price segmentation across metropolitan regions. Ensemble methods such as random forests and gradient boosting extend this idea by combining multiple trees, improving predictive accuracy while preserving the ability to model interaction effects. This is the interpretability–performance trade-off central to applied housing analytics, and recent work shows that it can be addressed through feature importance measures and rule-based interpretation of tree splits [28, 34].

These properties make tree-based methods a suitable choice for the heterogeneous U.S. metropolitan housing data examined in this thesis. Here the aim is not prediction alone, but to identify the drivers of housing prices and the segmentation patterns across metropolitan regions.

2.5. Research Gap and Contribution of the Thesis

The previous sections highlight three complementary strands of literature. SDG research explains how sustainability outcomes emerge through interdependencies, synergies, and trade-offs rather than isolated indicators [26]. Housing research documents the structural drivers of affordability and the strong regional heterogeneity of U.S. metropolitan markets [12]. Empirical work shows that tree-based machine learning improves the analysis of non-linear housing price dynamics relative to linear approaches [28].

Despite these contributions, an important gap remains at the intersection of these literatures. Broad sustainability frameworks are often difficult to translate into concrete and measurable urban domains. Within housing analysis, affordability, rental pressure, and market competitiveness are frequently examined separately rather than within a single analytical framework. At the same time, machine learning applications often prioritise predictive accuracy while giving less attention to interpretability, which reduces their value for decision-oriented analysis [17]. As a result, the connection between sustainability framing, housing market analysis, and machine learning remains underdeveloped.

This thesis addresses that gap in three ways. First, it narrows the discussion to a concrete urban domain by treating U.S. housing markets as an empirical setting through which SDG 11-related challenges can be examined. Second, it combines affordability, rental pressure, and market competitiveness indicators within a single data-driven framework. Third, it applies interpretable tree-based machine learning not only to predict housing prices but also to explain how affordability-related factors shape housing price segmentation across metropolitan regions.

3. Working Methodology

3.1. Data Sources and Collection Methods

3.1.1. Zillow Housing Data Sources

The dataset used in this thesis was obtained from **Zillow Research**, a public data platform that provides regularly updated information on the U.S. housing market. It includes variables such as median sale prices, rental prices, market heat indices, affordability metrics, and new construction prices. These indicators capture both how housing markets change over time and how conditions differ across regions, which makes them well suited to the temporal and regional focus of this study. Zillow Research was selected as the primary data source because of its broad geographic coverage and the consistency of its housing indicators across metropolitan areas.

Within this dataset, the analysis centres on affordability and market demand, as these variables are closely related to housing price dynamics. Tracking these indicators on a monthly basis from 2018 to 2025 allows both short-term fluctuations and longer-term trends in housing prices to be examined.

3.1.2. Data Integration and Structure

To enable structured time-series analysis, the seven datasets listed below were merged into a single dataset, with all variables aligned by region and date:

- Median Sale Price
- Percentage of Homes Sold Above List Price
- Median Rent (ZORI)
- Market Heat Index
- Affordability (Years to Save)
- Sale-to-List Price Ratio
- New Construction Median Sale Price

Each dataset was reshaped into long format in Python and merged using common identifiers: **RegionID**, **RegionName**, **RegionType**, **StateName**, and **Date**. The final dataset comprises 10,500 rows and 12 columns, covering the period from **March 2018 to February 2025**.

Variable Descriptions:

- **RegionID:** Unique code for each geographic region.
- **RegionName:** Name of the city or metropolitan area.
- **RegionType:** Geographic level of the region; all records in this dataset are metropolitan statistical areas (MSA).
- **StateName:** U.S. state where the region is located.
- **Date:** Monthly timestamp of the observation.
- **MedianSalePrice:** Typical home sale price in the region.
- **PctSoldAboveList:** Percentage of homes sold above listing price, indicating market competitiveness.
- **MedianRent:** Typical rent in the region (Zillow Observed Rent Index, ZORI).
- **MarketHeatIndex:** Zillow's measure of supply-and-demand balance, where higher values indicate stronger demand relative to supply.
- **YearsToSave:** Estimated number of years a household would need to save for a down payment.
- **SaleToListRatio:** Ratio of sale price to original list price, showing pricing dynamics.
- **NewConMedianSalePrice:** Median sale price of newly constructed homes in the region.

3.2. Data Preprocessing and Cleaning

To ensure data quality and consistency, several preprocessing steps were applied to the merged dataset, including handling missing values, checking data types, detecting duplicates, and inspecting potential outliers. Missing values and duplicate entries were identified using Python.

The dataset initially contained 10,500 observations and 12 variables, where each row represents housing data for a specific region and month. The highest proportion of missing data was observed in **NewConMedianSalePrice** (1,718 observations, approximately 16%). This variable was retained, as the missingness reflects the absence of new construction activity rather than data errors.

Smaller amounts of missing data in variables such as **PctSoldAboveList**, **MedianRent**, and **SaleToListRatio** were handled by removing only the affected rows (Figure 3.1).

A heatmap was used to visualise the distribution of missing values across variables. Basic data type checks were performed in Python to ensure correct formatting. No duplicate records were detected. After the cleaning process, the dataset was reduced to 9,893 observations (Figure 3.2).

```

Shape: (10500, 12)

Column Types:
RegionID          int64
RegionName        object
RegionType        object
StateName         object
Date              datetime64[ns]
MedianSalePrice   float64
PctSoldAboveList  float64
MedianRent        float64
MarketHeatIndex   float64
YearsToSave       float64
SaleToListRatio   float64
NewConMedianSalePrice float64
dtype: object

Missing values per column:
NewConMedianSalePrice  1718
PctSoldAboveList       301
SaleToListRatio        291
MedianRent             191
StateName              84
dtype: int64

```

Figure 3.1.: Data type and missing value summary for the merged dataset

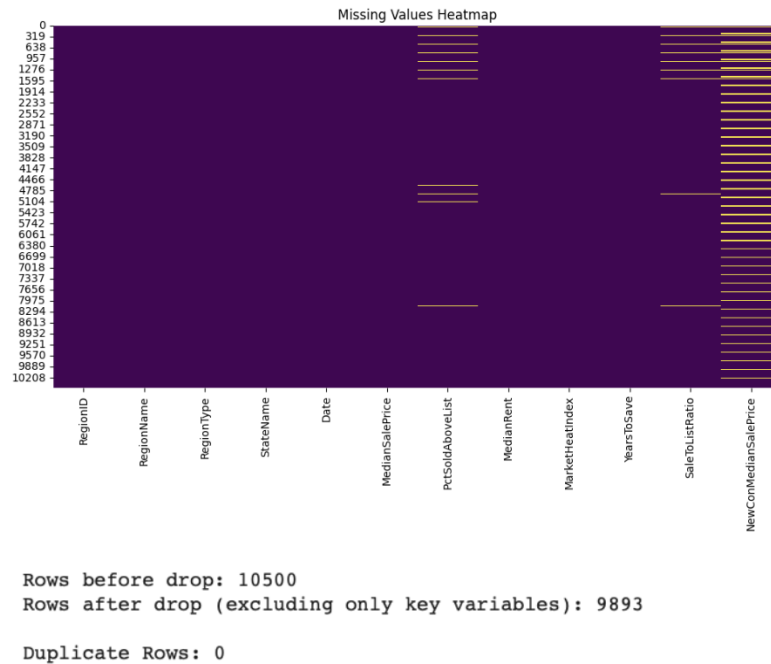


Figure 3.2.: Missing values heatmap and post-cleaning dataset statistics

Following the merging and cleaning processes performed in Python, the dataset was imported into SAS via the Manage Data interface. Additionally, the Discover Information Assets component was used to obtain a general overview of the dataset (Figure 3.3).

The dependent variable used in this analysis is **MedianSalePrice**, representing housing market value. Independent variables include **MedianRent**, **NewConMedianSalePrice**, **PctSoldAboveList**, **SaleToListRatio**, **YearsToSave**, and **MarketHeatIndex**, as they are considered influential factors in home prices. Summary statistics and variable distributions were further examined through the Column Analysis (Figure 3.4). The dataset comprises 9893 rows and 12 columns, with a total size of 940.2 KB. Out of these 12 columns, 8 are

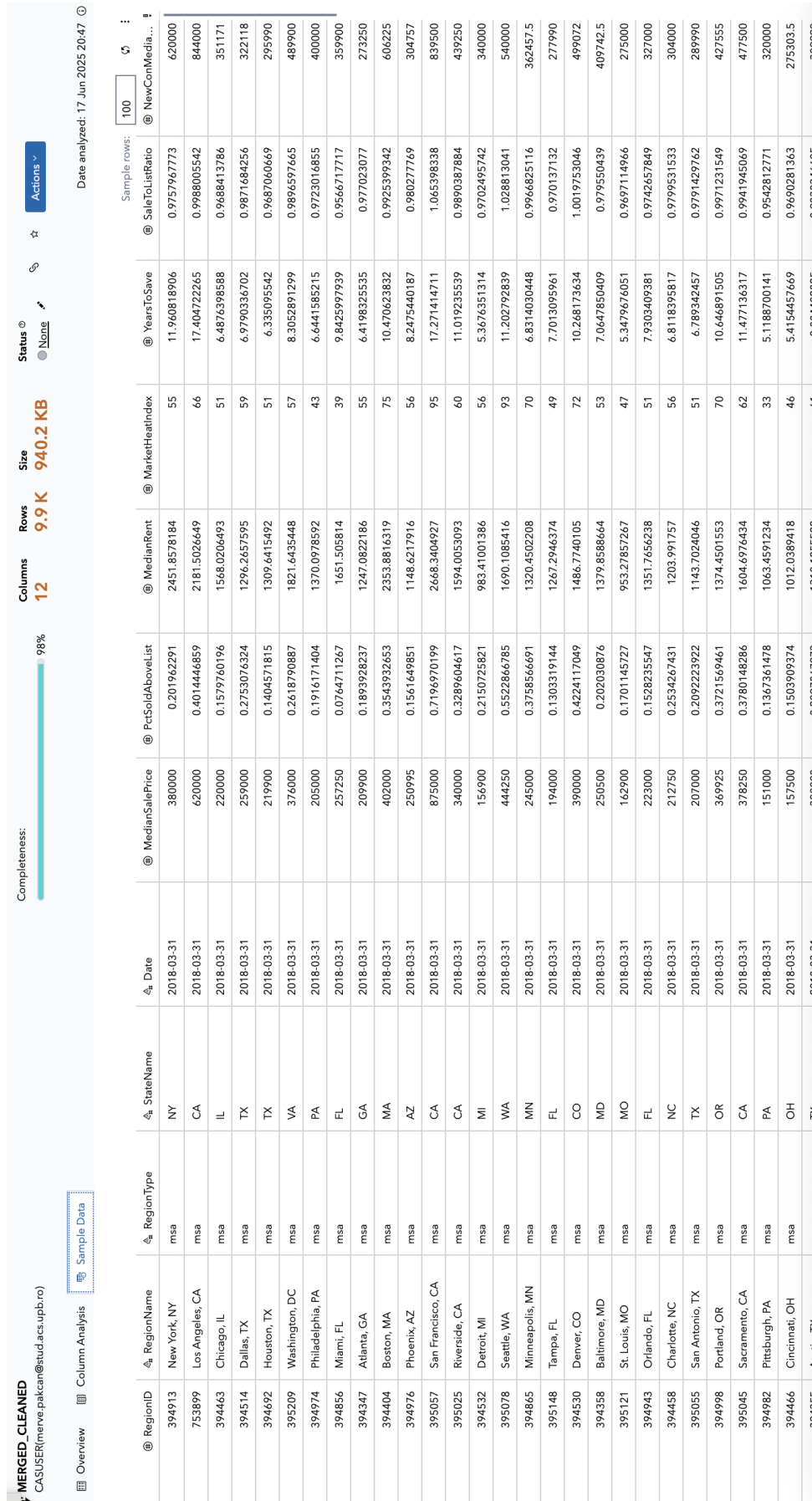


Figure 3.3.: Overview of the dataset in SAS Discover Information Assets

numerical (DataType: double) and 4 are character data (DataType: varchar), offering a suitable data structure for analysis.

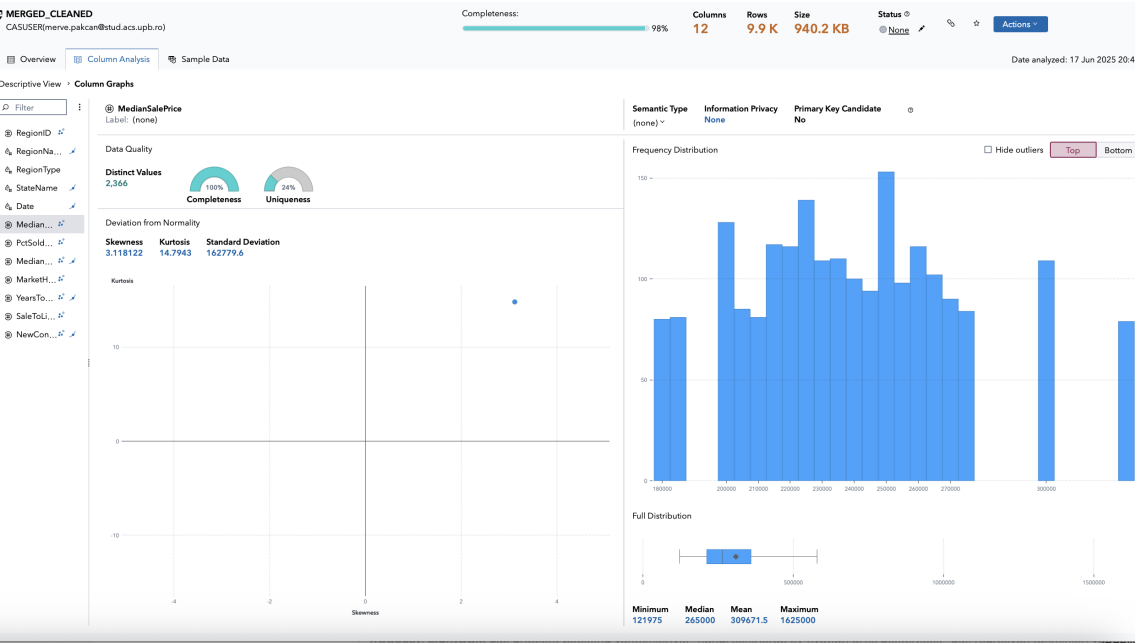


Figure 3.4.: Column analysis of the dataset

Also, after loading the data into the SAS environment, the variable types were verified in the Manage Data interface (Figure 3.5).

#	Name	Label	Data Type	Row Length	Formatted Length	Format
1	RegionID	--	double	8	12	--
2	RegionName	--	varchar	21	21	--
3	RegionType	--	varchar	3	3	--
4	StateName	--	varchar	2	2	--
5	Date	--	varchar	10	10	--
6	MedianSalePrice	--	double	8	12	--
7	PctSoldAboveList	--	double	8	12	--
8	MedianRent	--	double	8	12	--
9	MarketHeatIndex	--	double	8	12	--
10	YearsToSave	--	double	8	12	--
11	SaleToListRatio	--	double	8	12	--
12	NewConMedianSalePrice	--	double	8	12	--

Figure 3.5.: Data type summary of the dataset in SAS Data Explorer

Outlier detection was performed using the Explore and Visualise tools in SAS (Figure 3.6). Box plots were generated for key numerical variables to examine the distribution and identify potential outliers. The visualisations use gradient-shaded areas to highlight areas where observations are more concentrated, providing a clearer overview of variability across the housing indicators.

The variables `MedianSalePrice`, `NewConMedianSalePrice`, `PctSoldAboveList`, and `SaleToListRatio` show right-skewed distributions with clear upper outliers. Similar behaviour is observed for `YearsToSave` and `MarketHeatIndex`, indicating variation across regions. These outliers were particularly prominent in high-cost urban regions, reflecting real market conditions rather than anomalies.

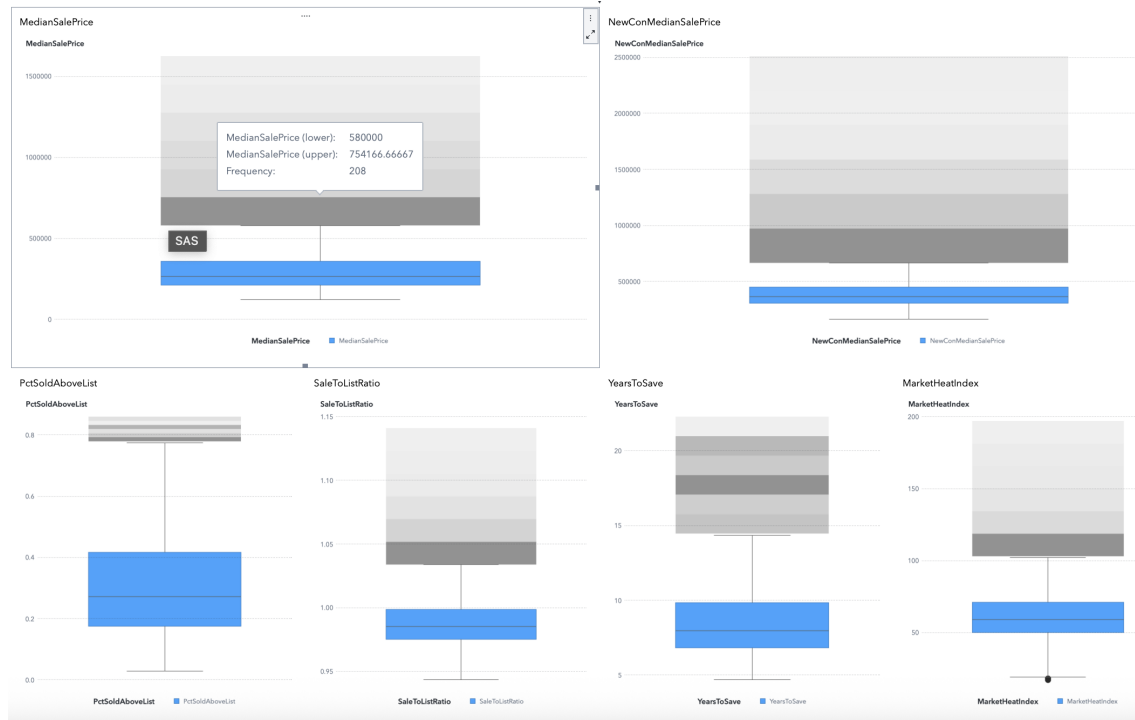


Figure 3.6.: Box plots for distribution and outlier analysis

The preprocessing and cleaning steps were carried out in Python, after which the cleaned dataset was imported into SAS Viya for the exploratory data analysis and subsequent modelling stages. This hybrid workflow combines Python’s flexibility for data manipulation with SAS Viya’s environment for visual analytics and machine learning.

3.3. Exploratory Data Analysis

Exploratory Data Analysis (EDA) serves as a foundational stage in data-driven research, aiming to uncover patterns, detect anomalies, and guide subsequent modelling through graphical and statistical techniques. More than a preliminary step, EDA plays a central role in understanding data structure and informing modelling decisions, as emphasised by Tukey [35].

3.3.1. Descriptive and Distribution Analysis

This section provides a general overview of the dataset through descriptive statistics and graphical analysis. Histograms and Q-Q plots are used to examine distribution patterns and normality, while potential outliers are identified to understand variability in the data.

Descriptive statistics provide a statistical overview of the main variables used in the analysis (Figure 3.7). The **MedianSalePrice** shows substantial variation across regions, with a mean of approximately \$309,000 and a maximum value exceeding \$1.6 million. New construction properties (**NewConMedianSalePrice**) reach up to \$2.5 million in some cities, indicating high-end market segments.

	RegionID	Date	MedianSalePrice	\
count	9893.000000	9893	9.893000e+03	
mean	406601.992621	2021-10-16 19:09:28.038006528	3.096715e+05	
min	394312.000000	2018-03-31 00:00:00	1.219750e+05	
25%	394531.000000	2020-01-31 00:00:00	2.131770e+05	
50%	394792.000000	2021-10-31 00:00:00	2.650000e+05	
75%	395005.000000	2023-07-31 00:00:00	3.599000e+05	
max	753912.000000	2025-02-28 00:00:00	1.625000e+06	
std	64112.991749	NaN	1.627796e+05	

	PctSoldAboveList	MedianRent	MarketHeatIndex	YearsToSave	\
count	9893.000000	9893.000000	9893.000000	9893.000000	
mean	0.306217	1500.371539	61.627211	8.631004	
min	0.029102	806.913837	17.000000	4.670567	
25%	0.175565	1208.192870	50.000000	6.822396	
50%	0.271981	1402.893622	59.000000	7.959976	
75%	0.416761	1715.942181	71.000000	9.836859	
max	0.859512	3347.407630	197.000000	22.269278	
std	0.163279	429.354211	17.170322	2.625767	

	SaleToListRatio	NewConMedianSalePrice
count	9893.000000	8.481000e+03
mean	0.988815	4.059965e+05
min	0.943317	1.620000e+05
25%	0.975242	3.062750e+05
50%	0.985311	3.650000e+05
75%	0.998708	4.499900e+05
max	1.140964	2.510000e+06
std	0.020732	1.759764e+05

Figure 3.7.: Descriptive statistics of the key variables in the dataset

The `PctSoldAboveList` variable, with an average of 31%, indicates that a considerable proportion of homes are sold above the listing price.

Rental prices (`MedianRent`) range from \$806 to over \$3,300, while the `MarketHeatIndex` spans from 17 to 197, highlighting differences in demand and buyer activity across regions.

`YearsToSave`, which estimates the number of years required to save for a down payment, averages at 8.6 years and reaches beyond 22 years in some areas. This reflects affordability challenges across cities.

The distribution of key numerical variables was examined using histograms generated in the SAS Visual Analytics environment (Figure 3.8). These plots provided a clear visual summary of the frequency and skewness patterns across the dataset.

Variables such as `MedianSalePrice`, `NewConMedianSalePrice`, and `MedianRent` exhibit right-skewed distributions. This is consistent with real-world housing markets, where a majority of properties cluster around mid-range values, but a smaller portion of high-value listings extend the upper tail. These skewed patterns reflect the wide range of housing prices across regions.

Conversely, `MarketHeatIndex` followed a near-normal distribution, indicating that most markets experience moderate buyer competition, with fewer cases showing exceptionally high or low heat levels. `PctSoldAboveList` and `SaleToListRatio` also revealed slight right skewness, supporting the presence of competitive market conditions where homes often sell above their listing price.

`YearsToSave` showed a right-skewed profile as well, with the majority of regions requiring a moderate savings period, while a smaller number of areas, typically high-cost urban centres, require significantly longer timeframes to afford a median-priced home.

Q-Q plots were used to assess the normality of key numerical variables in the housing dataset (Figure 3.9). Most variables, including `MedianSalePrice`, `MedianRent`, and

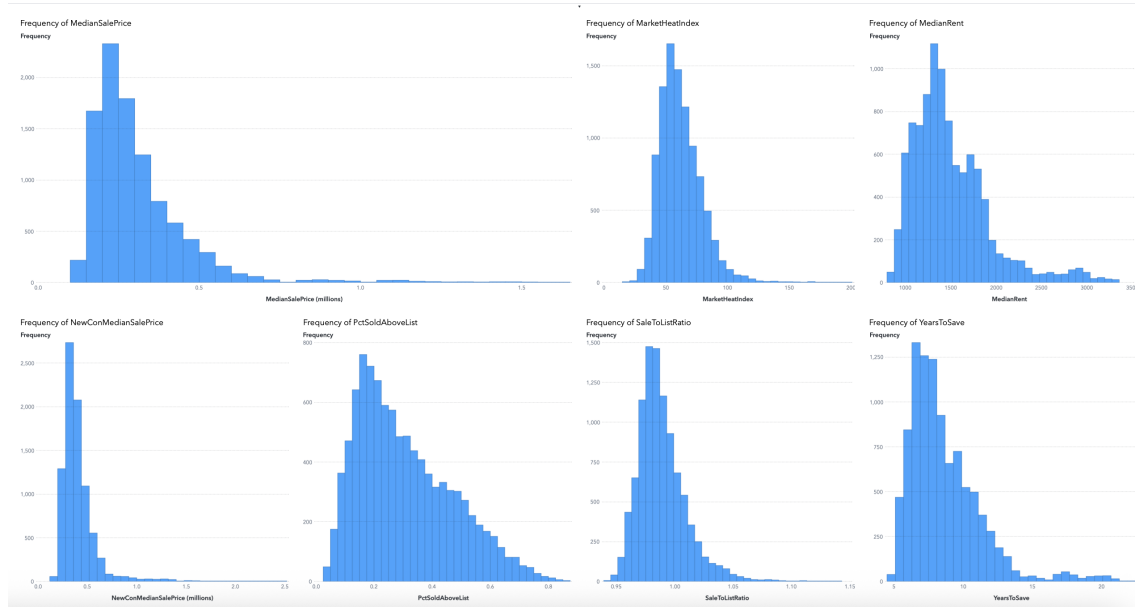


Figure 3.8.: Histograms of key housing market variables

YearsToSave, display clear deviations from the theoretical reference line, particularly at the tails. This indicates strong right-skewness and the presence of extreme values.

For example, **MedianSalePrice** and **MedianRent** exhibit pronounced curvature at both ends of the distribution, reflecting a small number of regions with exceptionally high prices. Similarly, variables such as **PctSoldAboveList** and **MarketHeatIndex** show S-shaped or curved patterns, suggesting departures from normality and potential clustering.

These distributional patterns confirm that the normality assumption does not hold for most variables. This supports the use of methods that do not rely on assumptions of normality.

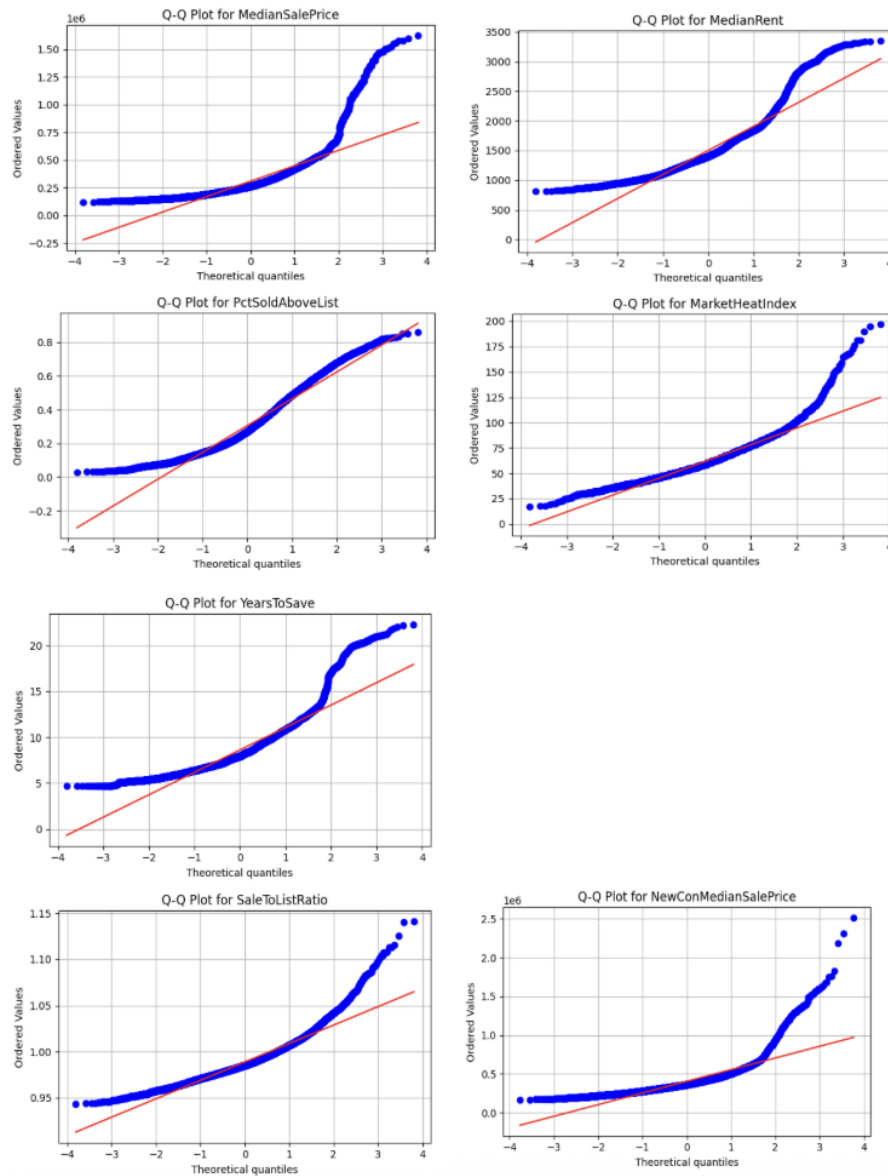


Figure 3.9.: Q-Q plots for normality assessment of numerical variables

3.3.2. Relationship Analysis

The correlation matrix, visualised in Python, illustrates the strength and direction of linear relationships among the key housing indicators (Figure 3.10). `MedianSalePrice` is strongly and positively correlated with both `MedianRent` ($r = 0.85$) and `YearsToSave` ($r = 0.90$), indicating that higher home prices tend to coincide with higher rents and longer saving periods.

Strong association is found between `PctSoldAboveList` and `SaleToListRatio` ($r = 0.94$), reflecting market competitiveness. `MarketHeatIndex` is only weakly related to prices ($r = 0.21$ with `MedianRent` and 0.37 with `MedianSalePrice`) but more strongly related to the competitiveness measures `PctSoldAboveList` and `SaleToListRatio` ($r = 0.70$ for both), showing that it reflects market competition rather than price levels.

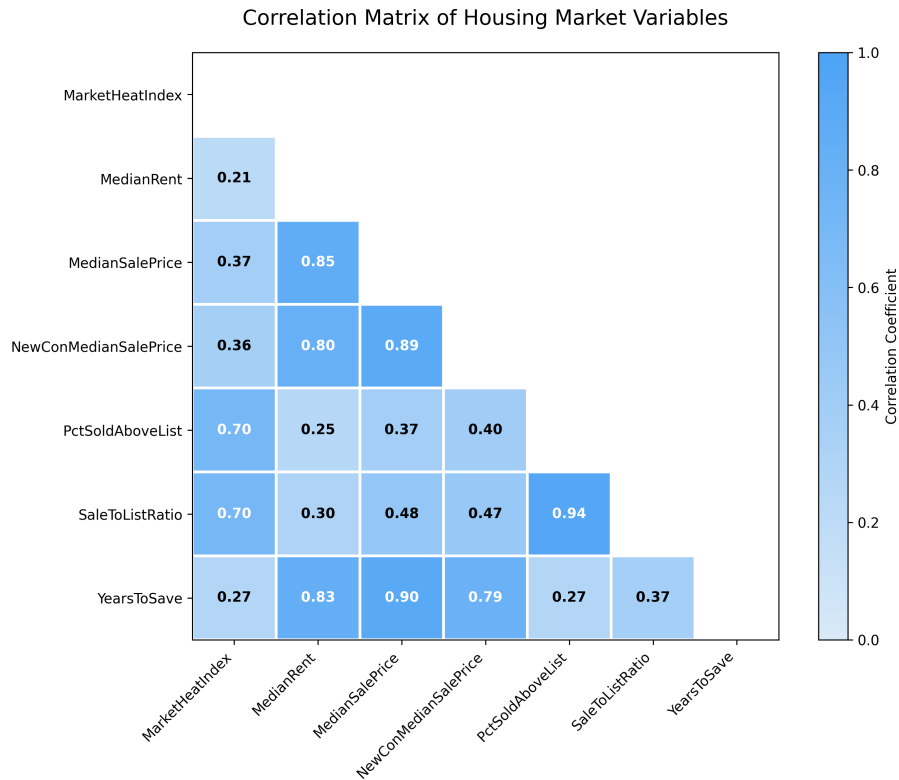


Figure 3.10.: Correlation matrix for relationship analysis

3.3.3. Temporal and Regional Analysis

To capture long-term trends in housing dynamics, time series plots were generated for average monthly values of median sale price and median rent in the U.S. between 2018 and early 2025 (Figure 3.11). **MedianSalePrice** increased steadily over time. The sharpest increase is observed between 2021 and 2022, followed by fluctuations at a higher price level.

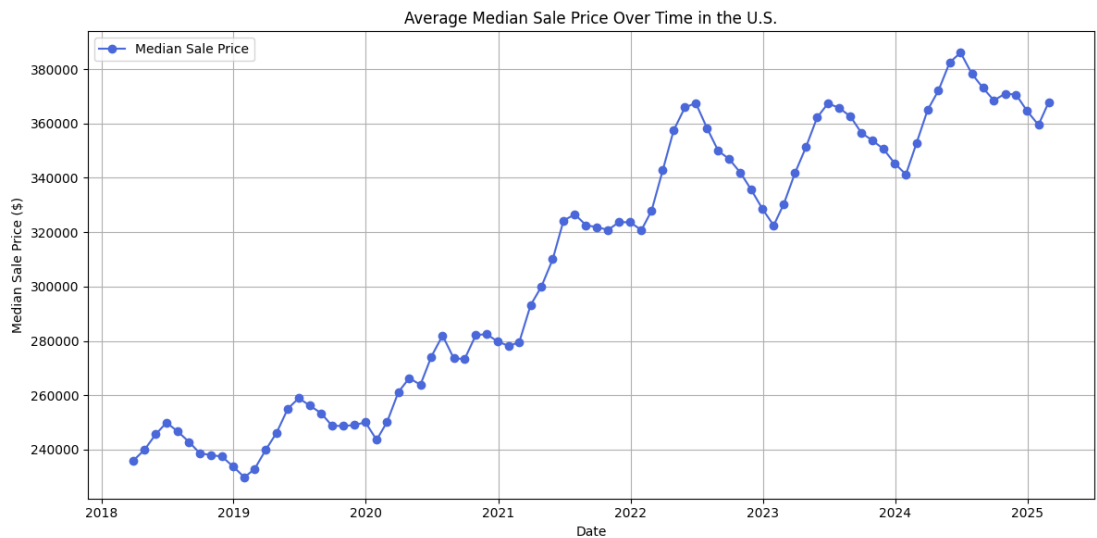


Figure 3.11.: Average monthly median sale price over time

MedianRent showed a similar pattern, increasing gradually before rising more sharply from late 2021 onwards (Figure 3.12). Together, the upward trends in both sale prices and rents indicate increasing affordability pressures over the study period. This observation is consistent with the findings of Glaeser and Gyourko [11], who associate worsening affordability conditions with persistent constraints in housing supply.

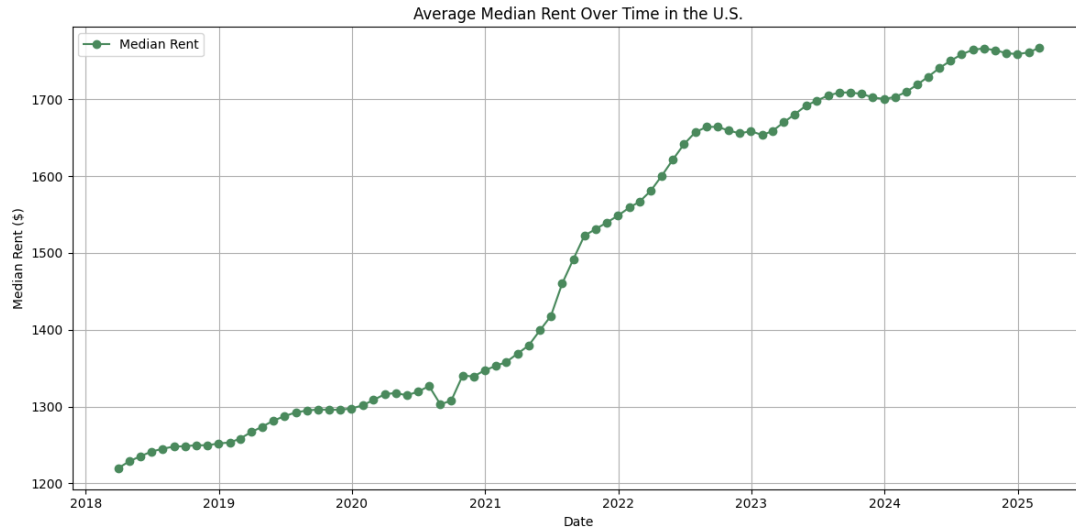
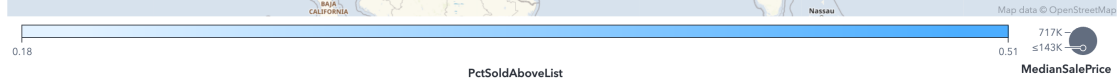
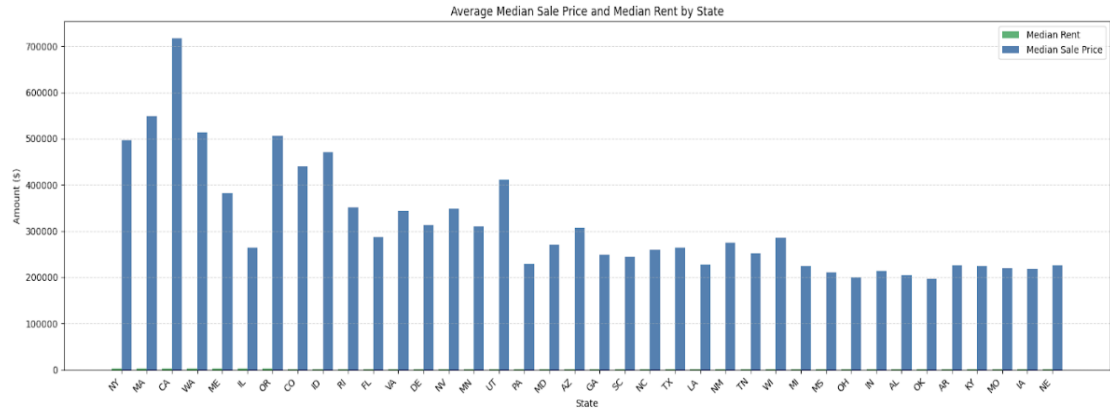


Figure 3.12.: Average monthly median rent over time

Regional variations in median sale prices and rental values across states are examined. To better illustrate the substantial differences between median sale prices and rents across states, the data was sorted in descending order (Figure 3.13). This highlights the regional disparities more clearly.

California has the highest median home sale price, followed by Washington and Massachusetts, all of which also report relatively high rental costs. Notably, New York has the highest median rent but does not appear among the top three states for sale prices.

In states like Idaho and Utah, home prices are comparatively high while rents remain moderate, resulting in a significant gap between ownership and rental markets. In contrast, Southern states such as Oklahoma and Mississippi exhibit both low sale prices and low rents, reflecting more affordable housing conditions. The Geo Bubble Map provides a spatial overview of the U.S. housing market by visualising **MedianSalePrice** (bubble size) and the **PctSoldAboveList** ratio (colour intensity) across states. Larger bubbles indicate higher sale prices, while more saturated blue tones reflect stronger competition, with more homes sold above the listing price. As shown in Figure 3.14, California (CA) records by far the highest median sale price (around \$717K) alongside strong competition (about 49% sold above list). The most competitive markets, however, are Massachusetts (about 51%) and Wisconsin (about 50%), the latter reaching this level despite a much lower price level (around \$285K). This pattern suggests that high prices and strong competition often concentrate in coastal and metropolitan areas, though competitive markets can also emerge where prices remain moderate.



3.4. Feature Engineering

To enhance the analytical depth, two derived features were engineered from the original Zillow housing dataset using SAS Visual Analytics:

- **RentToPriceRatio:** Calculated as `MedianRent` divided by `MedianSalePrice`, this metric serves as a proxy for affordability and rental yield. It reflects the relative cost of renting versus owning a home.
- **MarketTension:** Constructed by multiplying `MarketHeatIndex` with `PctSoldAboveList`, this composite indicator captures latent demand and market competitiveness, highlighting regions with stronger upward pricing pressures.

3.5. Modelling Approach

3.5.1. Preliminary Model Comparison

Before committing to a tree-based modelling strategy, a preliminary comparative analysis was carried out in SAS Visual Analytics to assess the relative performance of several regression-oriented methods on the prepared housing dataset. Four candidate models were evaluated using `MedianSalePrice` as the target variable: linear regression, a single decision tree, a random forest, and a gradient boosting model.

Linear regression provided a clear baseline, but its residual diagnostics showed a pronounced fan-shaped pattern consistent with strong heteroscedasticity and unmodelled non-linearity, confirming that a purely linear specification was insufficient for this dataset. The single decision tree, grown with default settings, produced affordability-driven splits led by `YearsToSave` and `RentToPriceRatio`, yet exhibited signs of underfitting typical of a shallow single tree and did not outperform the linear baseline in terms of Average Squared Error (ASE). The ensemble methods performed substantially better: the random forest achieved low out-of-bag error with stable training-validation convergence, while gradient boosting reached the lowest ASE among all four models, with error stabilising within the first few dozen boosting iterations.

As shown in Figure 3.15, the ensemble methods (Random Forest and Gradient Boosting) clearly outperform both the linear regression and the single decision tree on this dataset, while `YearsToSave` and `MedianRent` consistently emerge among the strongest predictors across all models. These findings motivated the focus on decision trees, random forests, and gradient boosting, which combine competitive accuracy with variable importance.

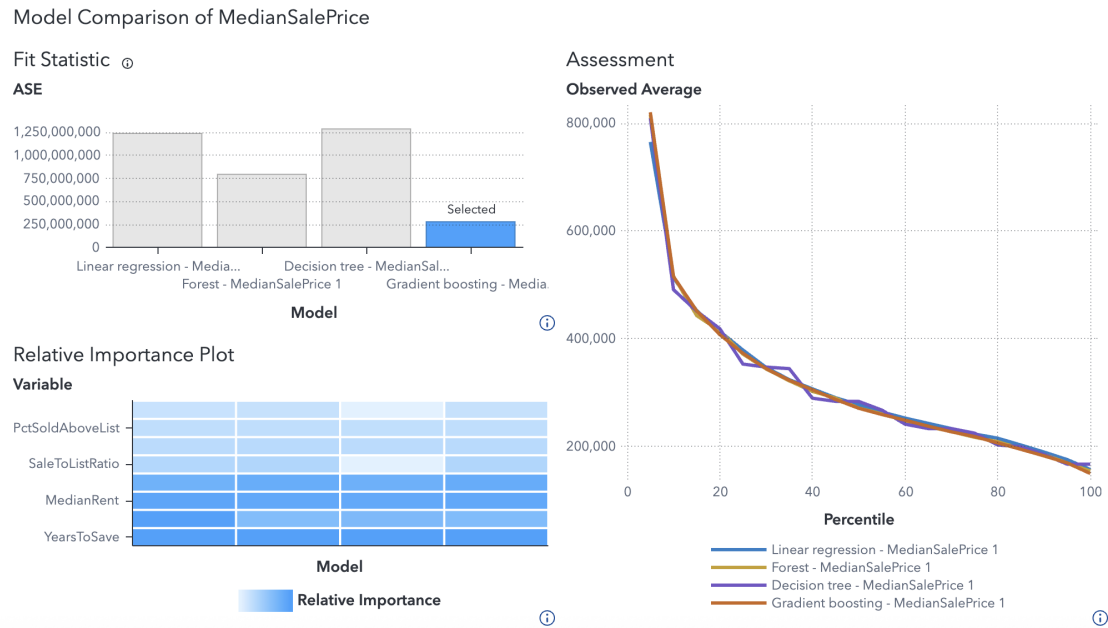


Figure 3.15.: Preliminary comparison of linear and tree-based models

3.5.2. Tree-Based Machine Learning Models

Urban housing markets exhibit strong nonlinearity and heterogeneity across regions. Housing prices are shaped by the combined effects of affordability, rental pressure, market competitiveness, and local economic conditions. Capturing these relationships is challenging for traditional statistical models, which often rely on restrictive assumptions regarding functional form and variable interactions. This limitation of parametric hedonic models has also been noted in recent real estate literature, which points to the restricted inferential capacity of traditional approaches when faced with complex market dynamics [17].

Tree-based machine learning methods offer a flexible and transparent framework for analysing such market structures. By recursively partitioning the input space using simple decision rules, decision trees can capture threshold effects and variable interactions present in the data. This makes them particularly suitable for exploratory housing market analysis, where understanding the drivers of price variation is as important as prediction accuracy. Beyond single-tree models, this family also includes ensemble methods such as random forests and gradient boosting, which combine multiple decision trees to improve predictive performance and robustness.

Within this modelling family, decision trees constitute the core analytical component. A decision tree organises the data into a hierarchical structure, where each split is based on a single explanatory variable and each terminal node represents observations with similar characteristics. As nested rule-based structures, these models translate data patterns into transparent if-else decision paths, thereby enabling straightforward interpretation of how key variables influence housing prices [17].

These models are implemented and compared in SAS Viya, which provides an integrated workflow for model development, tuning, and evaluation.

3.5.3. Decision Tree Algorithms

Split selection plays a key role in capturing variation in housing prices. Decision trees construct a hierarchical partition of the predictor space by recursively selecting splits that minimise within-node variability of the target variable. Since housing prices are continuous outcomes, this research uses regression trees, where splits are chosen based on variance reduction, also referred to as the residual sum of squares (RSS) or ANOVA criterion [32].

As the response variable is continuous, classification-based and statistical test-based splitting criteria such as Gini index, entropy, chi-square, or CHAID were not applicable and therefore not used. In SAS Viya, regression trees rely on binary splits and evaluate candidate thresholds across continuous predictors using this variance-based criterion, resulting in progressively more homogeneous price segments.

The first split occurs on `YearsToSave`, indicating that affordability-related factors explain a substantial share of the observed variation in housing prices.

In terms of tree pruning, classical post-pruning techniques, such as cost-complexity pruning, were not explicitly applied. Instead, tree complexity was controlled through pre-pruning constraints implemented via the decision tree hyperparameters available in SAS Viya.

Model growth was regulated using parameters such as maximum tree depth, minimum leaf size, and interval binning, which limit overly specific splits during training. These constraints helped reduce sensitivity to noise and supported more stable generalisation.

As the regression tree was designed to serve as an interpretable baseline model, pre-pruning was preferred in order to preserve clear decision rules and coherent segmentation structures. This modelling choice is consistent with prior housing market studies that use decision trees as interpretable tools for uncovering heterogeneous pricing patterns and attribute-based segmentation [8]. Additional regularisation and performance gains were subsequently obtained through ensemble methods.

Regression Trees vs. Classification Trees

Decision trees can be applied to both classification and regression problems, depending on the nature of the response variable (Figure 3.16). Classification trees are used when the outcome is categorical, whereas regression trees are designed for continuous response variables. Since housing prices are continuous, this analysis focuses on regression trees, which partition the predictor space using threshold-based splits, with each terminal node associated with the average price of that segment. This structure supports both prediction and meaningful analysis of housing price variation under different market conditions [32].

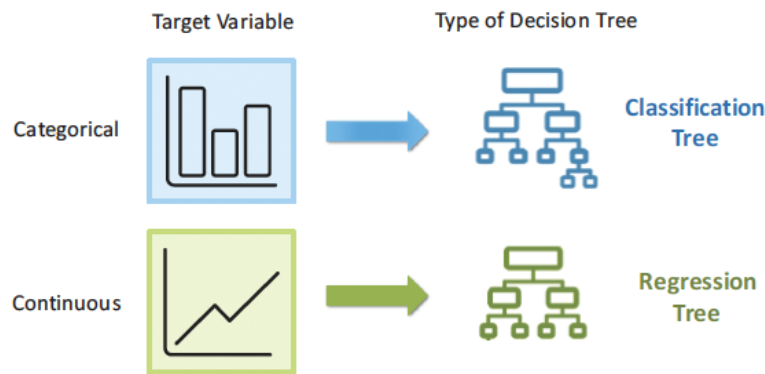


Figure 3.16.: Types of decision trees. Source: [32]

3.5.4. Ensemble Models

Bagging Versus Boosting

Ensemble learning methods enhance predictive performance by aggregating the outputs of multiple decision trees. Within this framework, bagging and boosting represent two distinct ensemble strategies that differ in their learning design and approaches to error reduction. Bagging trains multiple decision trees independently using bootstrap samples. The predictions of these trees are then combined through simple averaging. This strategy primarily targets variance reduction and leads to more stable model predictions. Boosting, in contrast, follows a sequential learning process. Each new tree is trained to correct the errors made by previous models. As a result, boosting focuses on reducing bias and often achieves higher predictive accuracy [13].

Given the heterogeneous structure of housing prices, both approaches are considered as candidate modelling approaches. Random Forest and Gradient Boosting are selected as representative implementations of bagging and boosting, respectively. Comparing these models allows an assessment of whether variance-focused aggregation or bias-reducing sequential learning is more effective for housing price prediction.

Table 3.1 provides a comparative summary of the conceptual differences between bagging and boosting, including sampling goals, learning structures, and representative algorithms.

Table 3.1.: Comparison of bagging and boosting methods (adapted from [32])

	Bagging	Boosting
Sampling	Randomly chosen unweighted cases	Higher weight for poorly predicted observations
Goal	Minimise variance	Increase accuracy (reduce bias)
Combining models	Weighted average	Weighted aggregation
Learning structure	Parallel trees	Sequential trees
Example algorithms	Random Forest	Gradient Boosting

Beyond these operational aspects, the structural divergence between the two methods is illustrated below. As shown in Figure 3.17, bagging constructs multiple decision

trees in parallel and aggregates their predictions, leading to reduced variance and more stable outcomes.

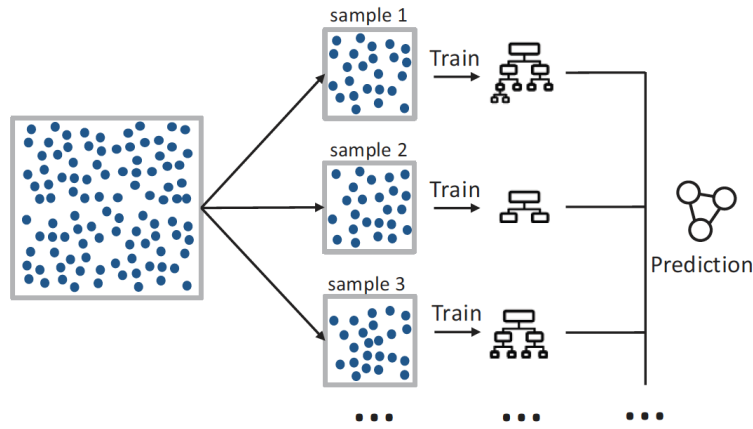


Figure 3.17.: Parallel ensemble approach. Source: [32]

Figure 3.18 highlights the sequential nature of boosting, where each tree iteratively refines the model by focusing on previously mispredicted observations. This sequential learning mechanism is particularly effective for capturing nonlinear relationships commonly observed in urban housing markets.

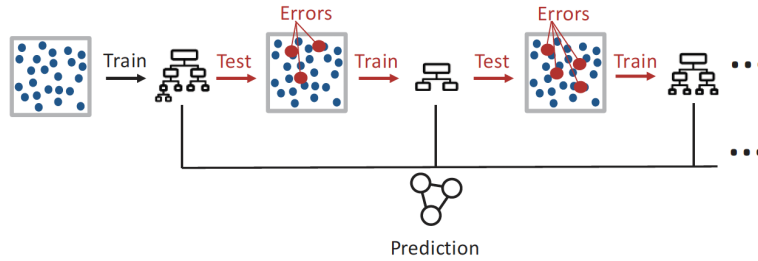


Figure 3.18.: Sequential ensemble approach. Source: [32]

From a bias–variance perspective, these two strategies address prediction error through different mechanisms. Evaluating both Random Forest and Gradient Boosting therefore enables a systematic assessment of which mechanism better captures housing price dynamics in a heterogeneous market environment.

3.6. Chapter summary

This chapter presented the methodological framework applied in this research. The dataset was sourced from Zillow Research, covering U.S. housing markets from 2018 to 2025. Seven individual datasets were merged and cleaned into a structured analytical dataset comprising 9,893 observations. Exploratory data analysis revealed right-skewed distributions across key variables, strong positive correlations between `MedianSalePrice`, `MedianRent`, and `YearsToSave`, as well as notable regional disparities concentrated in coastal and metropolitan areas. Two derived features were also introduced: `RentToPriceRa-`

`tio`, reflecting the relative cost of renting versus owning, and `MarketTension`, capturing demand intensity and competitive pricing pressure.

The modelling approach introduced decision trees as a transparent baseline, alongside Random Forest and Gradient Boosting as ensemble methods. A preliminary model comparison confirmed that ensemble approaches outperform both linear regression and single decision trees in capturing nonlinear housing price relationships. These findings provide the foundation for the model development, tuning, and evaluation discussed in Chapter 4.

4. Data Modelling and Results

This chapter implements and evaluates the modelling framework established in Chapter 3. A linear regression model is first developed as an interpretable baseline, followed by three tree-based approaches: decision trees, random forests, and gradient boosting. The analysis addresses both predictive performance and the structural drivers underlying housing price variation across U.S. metropolitan regions.

The three tree-based models used identical train–validation–test partitions, the same predictor set, and the full sample of 9,893 observations. A predefined partition variable divided the data into training, validation, and test subsets in an approximate 60/20/20 ratio (5,935, 1,979, and 1,979 observations, respectively). All models developed in this chapter predict the same target variable, `MedianSalePrice`, which allows their performance to be compared directly across the fit statistics and comparison tables that follow. The linear regression model is reported as an interpretable reference point rather than a strictly comparable competitor. Following the default behaviour of the Model Comparison node in SAS Model Studio, the champion model in each pipeline is selected based on the test partition.

Beyond error-based evaluation, the chapter emphasises interpretability, nonlinear relationships, and the segmentation patterns captured by tree-based methods. These are examined through visual analytics and model interpretation outputs.

4.1. Creating a Linear Regression Baseline

Linear regression was used as the baseline model before moving on to the tree-based approaches. It provides a reference point and a starting error level for comparison.

The baseline regression model was implemented using the Linear Regression object in the SAS Visual Analytics component. `MedianSalePrice` was defined as the response variable, while affordability, rental, and market competitiveness indicators were assigned as continuous effects. No automated variable selection procedure was applied, allowing all engineered and domain-relevant predictors to remain in the model for interpretability and direct comparison with the subsequent tree-based approaches.

The model was fitted in SAS Visual Analytics on `MedianSalePrice` using 8,481 of the 9,893 observations available. The remaining 1,412 records were excluded through listwise deletion, since linear regression cannot handle missing predictor values. Most of these records were missing the `NewConMedianSalePrice` value, which happens in regions that had no new construction during this period. This is not a data error, so the tree-based models keep these records and use the full sample instead.

Default modelling settings provided by SAS Viya were retained for the baseline specification. The confidence level was set to 0.95, and statistical significance was evaluated using a tolerance threshold of 10^{-10} . Assessment plots and residual diagnostics were enabled to evaluate model fit, prediction behaviour, and error structure across the prediction range. The purpose of the linear regression model was not optimisation, but rather to establish a transparent and interpretable benchmark against which the more flexible tree-based models could later be evaluated.

As shown in Figure 4.1, all predictors were statistically significant at $p < 0.0001$, with the strongest linear associations observed for **MedianRent**, **NewConMedianSalePrice**, **RentToPriceRatio**, and **PctSoldAboveList**. In the assessment curve, the predicted values closely followed the observed values across all percentiles. This shows that the model captures the overall shape of the price distribution, at least at a descriptive level.

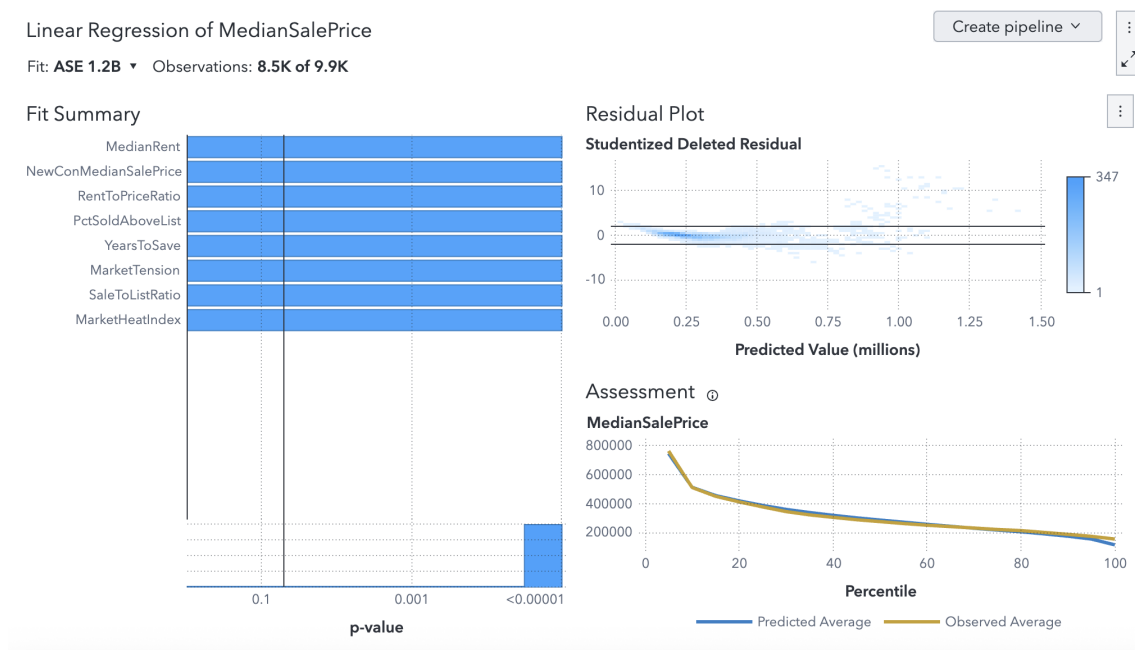


Figure 4.1.: Linear regression baseline model diagnostics and assessment plots

The residual plot, however, revealed a clear problem. For lower predicted prices, the residuals stayed close to zero, meaning the model predicted these prices accurately. As prices increased, the residuals spread out much more, exceeding ± 10 on the studentized residual scale for the most expensive homes. This means the model made larger errors for higher-priced markets. This pattern is known as heteroscedasticity: the error size grows as the price level grows. A linear model cannot handle this kind of changing error structure.

The model reached an Average Squared Error (ASE) of approximately 1.2×10^9 . This value is only a reference for the linear model itself. It is not directly comparable with the tree-based models in this chapter, because the linear regression used fewer observations (8,481) and included **RentToPriceRatio**, which is derived from the target variable. For this reason, the direct comparison of all models is made in Section 4.5, where every model uses the full 9,893 observations and the same predictor set.

Overall, linear regression served its purpose as a starting point. It identified which predictors are most strongly linked to housing prices and provided a clear error level for

comparison. However, its main limitation was the strong heteroscedasticity seen in the residual plot. This indicated that a linear model cannot fully capture the price structure, which motivated the use of the more flexible tree-based models.

`RentToPriceRatio` and `MarketTension` were used in the exploratory analysis, the preliminary comparison, and this linear regression baseline. They were not carried over to the tree-based models that follow. `RentToPriceRatio` was excluded, despite the strong association noted above, because it is derived from the target variable (`MedianSalePrice`) and could cause data leakage. `MarketTension` was excluded because tree-based models capture variable interactions on their own, making a separate interaction feature unnecessary.

4.2. Creating a Decision Tree Model

4.2.1. Regression Tree Interpretation and Visualisation

The regression decision tree is the first tree-based model considered in this analysis. This model is selected due to its ability to capture nonlinear interactions while keeping a clear, rule-based structure. No hyperparameter tuning was applied at this stage, as interpretability and rule extraction were prioritised over predictive optimisation. For the regression decision tree, the Explore and Visualize interface in SAS Visual Analytics was also used to inspect the splitting structure, variable importance, node rules, and diagnostic outputs (Figure 4.2) before the model was formalised as a pipeline in Model Studio. These visual checks clarified how the tree partitions the data and helped guide the model configuration. Both views use the same partition variable, so they describe the same model: one used for visual interpretation and the other for formal evaluation.

In Section 3.5.1, the decision tree was included in a preliminary comparison alongside linear regression and the ensemble models, in order to assess the candidate models at a high level. In this section, the regression tree is developed and analysed in detail for the first time, as the first tree-based model in the formal workflow. This establishes the regression tree as an interpretable reference point for the ensemble methods that follow.

The root node represents the entire dataset, and the first split is performed on `YearsToSave`. This means that `YearsToSave` reduces the prediction error more than any other variable, making it the most influential variable in the model.

As the tree grows downward, further splits are made using variables such as `MedianRent` and `NewConMedianSalePrice`. `YearsToSave` also appears again at lower levels, which reinforces its strong influence on median sale prices. The repeated use of these variables shows that affordability and rental pressure play the main role in explaining price differences. The thickness of each branch reflects the number of observations passing through that split, so thicker branches correspond to larger groups of data.

At the bottom of the tree, each leaf node corresponds to a group of observations with similar characteristics, and the model assigns it a single predicted price equal to the segment average. In Figure 4.2, these predictions are shown with a colour gradient, where darker shades mark higher-priced segments. The result is a clear partition of the housing

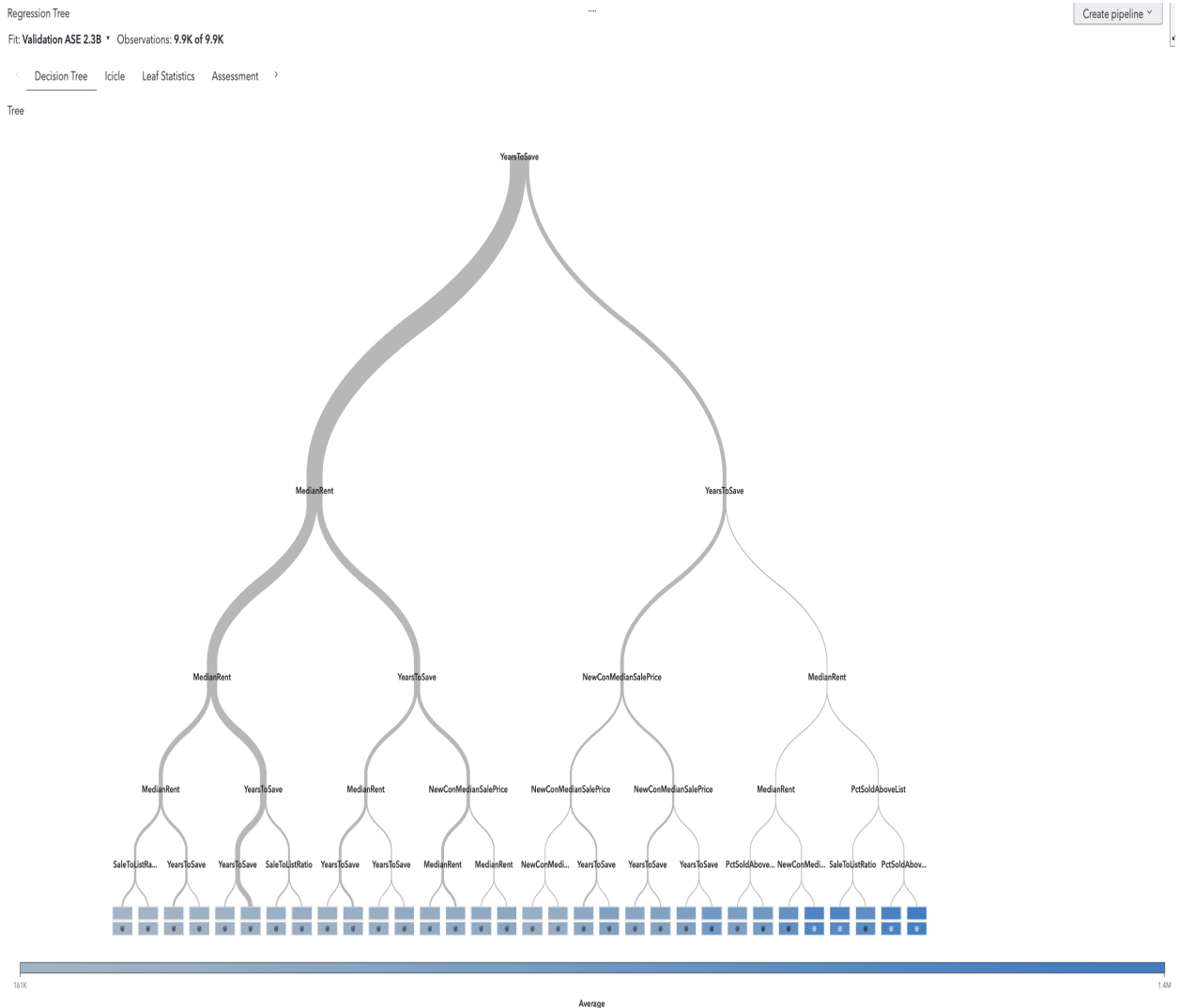


Figure 4.2.: Regression Tree structure for model interpretation

market into distinct price groups, from low-priced segments on one side to high-priced ones on the other.

After fitting the model on 9.9K observations, the regression tree yields an ASE of approximately 2.3B. This matches the validation RMSE of about 48K for the default Decision Tree later. Beyond predictive accuracy, the model is primarily informative due to the interpretability provided by its hierarchical splitting structure.

Variable importance results show a dominant contribution from **YearsToSave**, with **MedianRent** and **NewConMedianSalePrice** following at a secondary level (Figure 4.3). The market competitiveness indicators (**PctSoldAboveList**, **SaleToListRatio**, and **MarketHeatIndex**) contribute only marginally. This suggests that the tree separates housing markets into price groups mainly according to long-term affordability conditions rather than short-term buyer competition.

The icicle visualisation provides an alternative view of the regression tree, illustrating the hierarchical structure of successive splits (Figure 4.4). Large blocks for **YearsToSave**

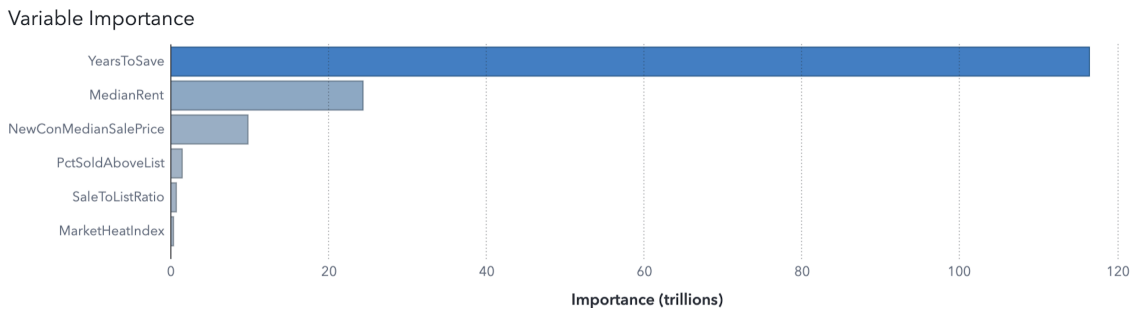


Figure 4.3.: Variable importance for identifying key predictors

and **MedianRent** dominate the upper levels, indicating that affordability and rental pressure variables drive the primary segmentation of the market. Competitiveness indicators such as **SaleToListRatio** and **PctSoldAboveList** appear only in the lower layers, refining distinctions within already-formed price segments. This pattern is consistent with the variable importance results and reinforces the interpretability of the regression tree.

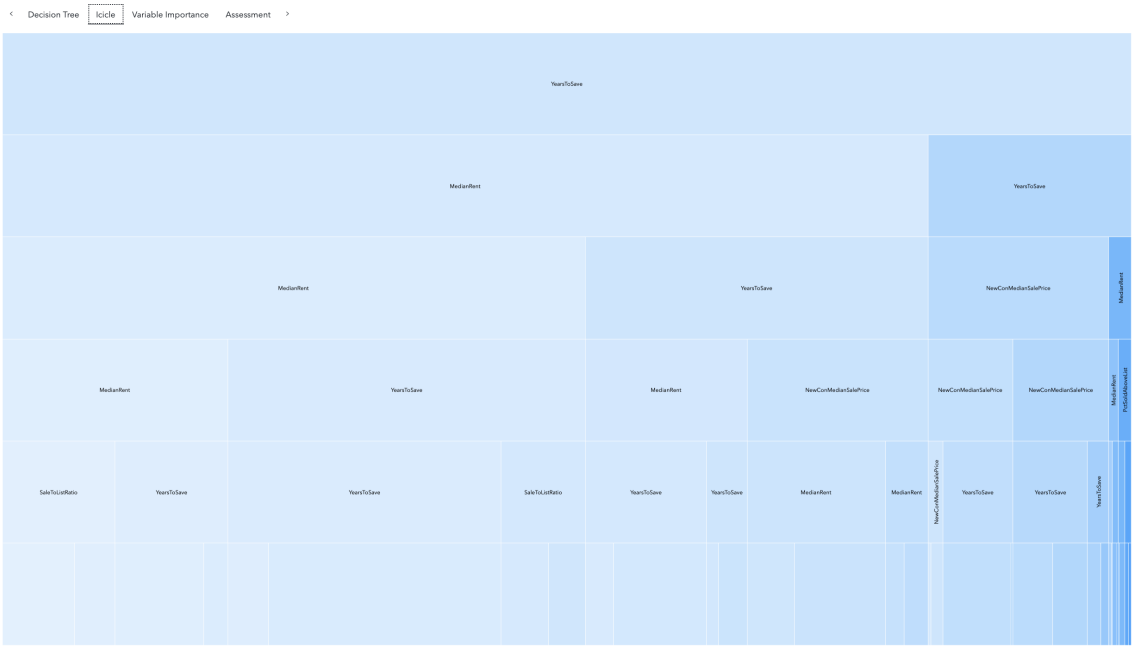


Figure 4.4.: Icicle plot for hierarchical split visualisation

The leaf statistics plot summarises the final segmentation produced by the regression tree (Figure 4.5). Each leaf corresponds to a group of observations sharing similar market characteristics, and the associated bar reports the average predicted median sale price for that cluster. The ordering of leaf means reveals a clear price hierarchy, with lower-priced clusters on the left and progressively higher-priced clusters on the right. This indicates that the model formed coherent and well-separated price segments, validating the segmentation structure learned by the tree.

The node rules output summarises how the regression tree partitions observations based on variable thresholds (Figure 4.6). Each row corresponds to a node and displays the splitting rules that group observations into progressively more homogeneous groups.

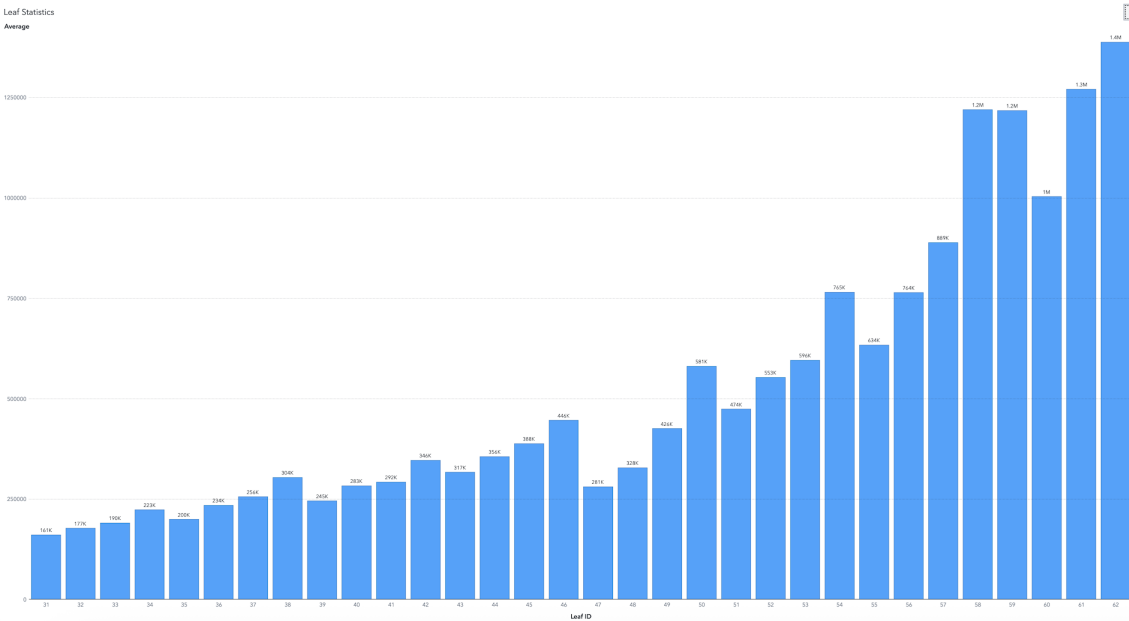


Figure 4.5.: Leaf statistics for terminal node comparison

The table gives a clear overview of these decision rules and the threshold values that define each split.

Although the input dataset does not contain missing values, SAS includes “missing” in the displayed rules as part of its default rule representation.

Node Statistics	Node Rules	Variable Importance	Cost-Complexity Pruning	Assessment	Assessment Statistics	Assessment Summary		
Node ID	Parent ID	Type	MarketHeatIndex	MedianRent	NewComMedianSalePrice	PctSoldAboveList	SaleToListRatio	YearsToSell
0	-1	Class						< 10.638 or Missing
1	0	Class						≥ 10.638
2	0	Class						< 10.638 or Missing
3	1	Class		< 1,418,064 or Missing				< 10.638 or Missing
4	1	Class		≥ 1,418,064				< 10.638 or Missing
5	2	Class						10.638 to 17.353
6	2	Class						≥ 17.353
7	3	Class		< 1,161,597				< 10.638 or Missing
8	3	Class		1,161,597 to 1,418,064 or Missing				< 10.638 or Missing
9	4	Class		≥ 1,418,064				< 8.765

Figure 4.6.: Node rules for regression tree segmentation

The residual plot (Figure 4.7) shows the typical pattern of a regression tree. Residuals appear in vertical bands because each terminal node produces a constant prediction. The residuals are centered around zero, so the model does not systematically over- or under-predict prices. They spread out at higher price levels, reflecting the greater price variability typically observed in expensive housing markets.

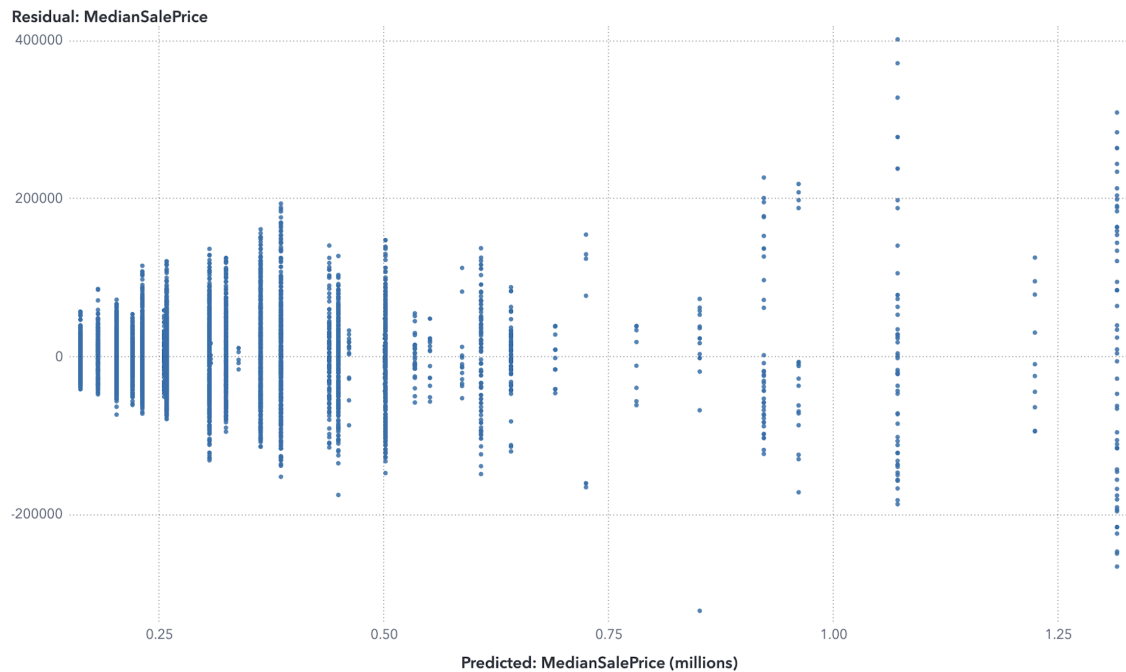


Figure 4.7.: Residual plot for error analysis

4.2.2. Default Decision Tree

The regression tree was then transferred to Model Studio as a pipeline to organise the workflow. The pipeline combines data preparation, decision tree training, and model evaluation in a single structure (Figure 4.8). This setup also makes it easier to add ensemble models in the next sections while keeping the evaluation approach consistent.

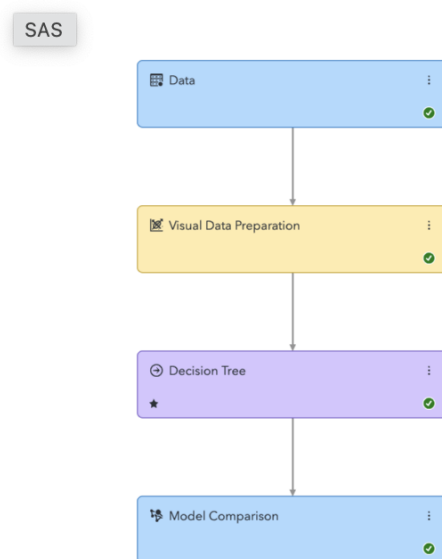


Figure 4.8.: Decision Tree modelling pipeline in SAS Model Studio

The regression tree demonstrates consistent performance across all data partitions, with an R^2 of 0.934 (training), 0.912 (validation), and 0.918 (test). The RMSE increases from approximately 41,300 dollars on the training set to 48,300 dollars on the test set, indicating a moderate increase in prediction error when moving from training to unseen data (Table 4.1). The remaining error suggests that more flexible methods could improve accuracy further. The regression tree therefore serves as the initial tree-based reference model for the ensemble methods evaluated next.

Table 4.1.: Fit statistics for the default Decision Tree model

Partition	N	RMSE (USD)	MAE (USD)	R^2
Train	5,935	41,290	29,974	0.934
Validation	1,979	48,065	32,100	0.912
Test	1,979	48,339	32,159	0.918

The predicted reports plot shows that most of the price differentiation is captured within the first few splits of the tree, after which the changes become more gradual (Figure 4.9). The training, validation, and test series remain closely aligned, indicating stable generalisation across data partitions without pronounced overfitting.

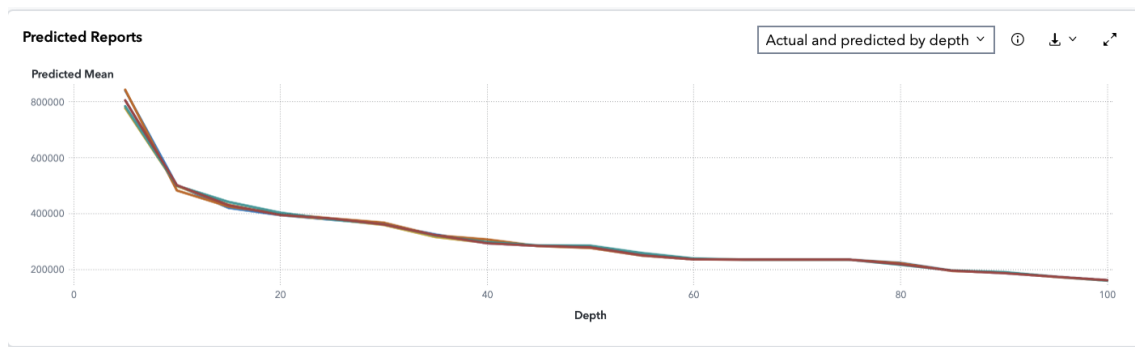


Figure 4.9.: Predicted mean by tree depth for evaluating generalisation across data partitions

4.2.3. Tuned Decision Tree

This pipeline was built in SAS Model Studio (Figure 4.10). Hyperparameter tuning was applied to the decision tree to improve its predictive performance while preserving its interpretability. Rather than modifying all available parameters, tuning was limited to a small set of hyperparameters that directly influence model complexity and generalisation.

The maximum tree depth was increased from 5 to 10 to better capture non-linear relationships in housing price dynamics that were not adequately represented in the default configuration. In parallel, the minimum leaf size was increased from 5 to 30 to limit overly specific terminal nodes and reduce sensitivity to noise. The number of interval bins was increased from 50 to 100 to allow finer splits for continuous variables. Other structural settings, including the variance-based splitting criterion and binary branching, were kept unchanged to preserve interpretability and ensure methodological consistency.

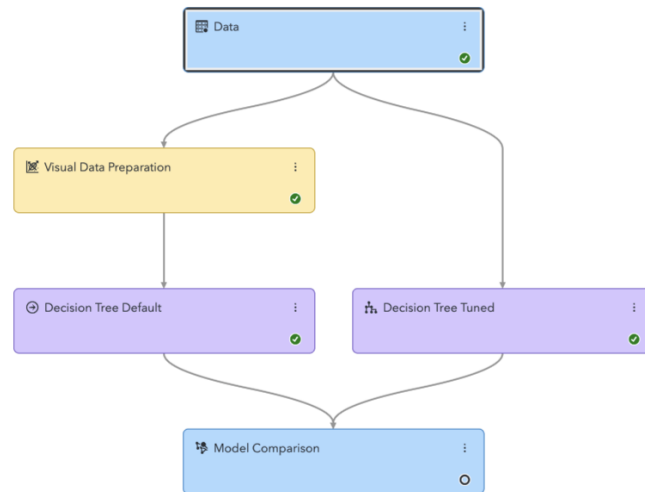


Figure 4.10.: Tuned Decision Tree Modelling Pipeline

Compared to the default decision tree, the tuned model achieves a clear improvement in predictive performance across all data partitions. On the test set, the RMSE decreased from about 48.3K to 39.0K, while the R^2 value increased from 0.918 to 0.947, indicating that the tuned tree captures housing price patterns more effectively (Table 4.2).

Table 4.2.: Fit statistics for the tuned Decision Tree model

Partition	N	RMSE (USD)	MAE (USD)	R^2
Train	5,935	35,322	22,777	0.952
Validation	1,979	40,019	25,550	0.939
Test	1,979	38,999	25,416	0.947

The comparison shows that the tuned decision tree achieves a lower RMSE and a higher R^2 than the default model on the test set (Table 4.3). The improvement comes mainly from the deeper splits, which increase model capacity. At the same time, the larger minimum leaf size limits overly specific nodes and keeps the complexity under control. As a result, the tuned tree reduces underfitting without causing overfitting and provides more stable generalisation than the default model.

Table 4.3.: Decision Tree model comparison on the test partition

Champion	Configuration	N	RMSE (USD)	R^2
★	Tuned	1,979	38,999	0.947
	Default	1,979	48,339	0.918

4.3. Creating a Random Forest Model

Random Forest modelling was implemented using two configurations: a default ensemble and a tuned ensemble. As a bagging-based method, random forests combine bootstrap aggregation with random feature selection, which reduces variance relative to the single regression tree. The default configuration serves as the reference ensemble, while the tuned model adjusts key hyperparameters to improve predictive accuracy and control overfitting. Both models were trained and evaluated within the same pipeline, ensuring methodological consistency and allowing a direct comparison under identical train-validation-test partitions (Figure 4.11).

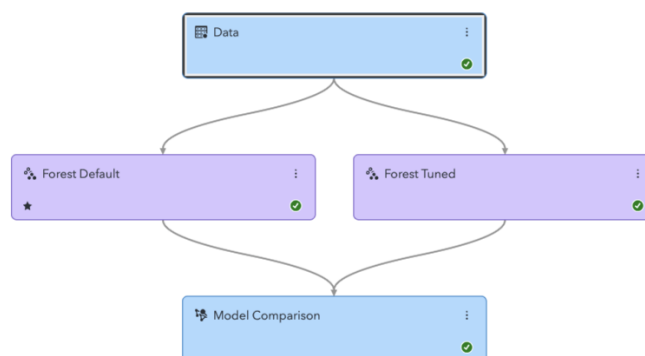


Figure 4.11.: Random Forest Modelling Pipeline

4.3.1. Default Random Forest

A default Random Forest was first trained using the SAS Viya Forest node without hyperparameter tuning. This baseline configuration represents a standard bagging-based ensemble and provides a benchmark for evaluating the incremental value of tuning and boosting in subsequent models.

In the default configuration, the Random Forest was trained with 100 trees, a maximum depth of 20, a minimum leaf size of 5, and approximately three variables considered at each split. The bootstrap sampling rate was set to 0.60. This configuration aggregates moderately deep trees to improve predictive accuracy while reducing the variance associated with a single decision tree.

The default Random Forest exhibited strong predictive performance across all partitions, achieving an R^2 of 0.984 on the training set, 0.971 on the validation set, and 0.971 on the test set (Table 4.4). The RMSE reduced substantially relative to the regression tree, decreasing from approximately 48K dollars to 28K dollars on the test set. This improvement suggests that the ensemble captures complex non-linear relationships more effectively than a single decision tree.

The error plot confirms this performance, showing a rapid decrease in the first trees followed by stabilization across all four curves (Figure 4.12). The train, validation, test,

Table 4.4.: Fit statistics for the default Random Forest model

Partition	N	RMSE (USD)	MAE (USD)	R^2
Train	5,935	20,687	13,006	0.984
Validation	1,979	27,721	17,166	0.971
Test	1,979	28,869	17,538	0.971

and out-of-bag curves run in parallel without diverging, indicating stable generalisation and diminishing returns beyond approximately 30 trees.

Error Plot

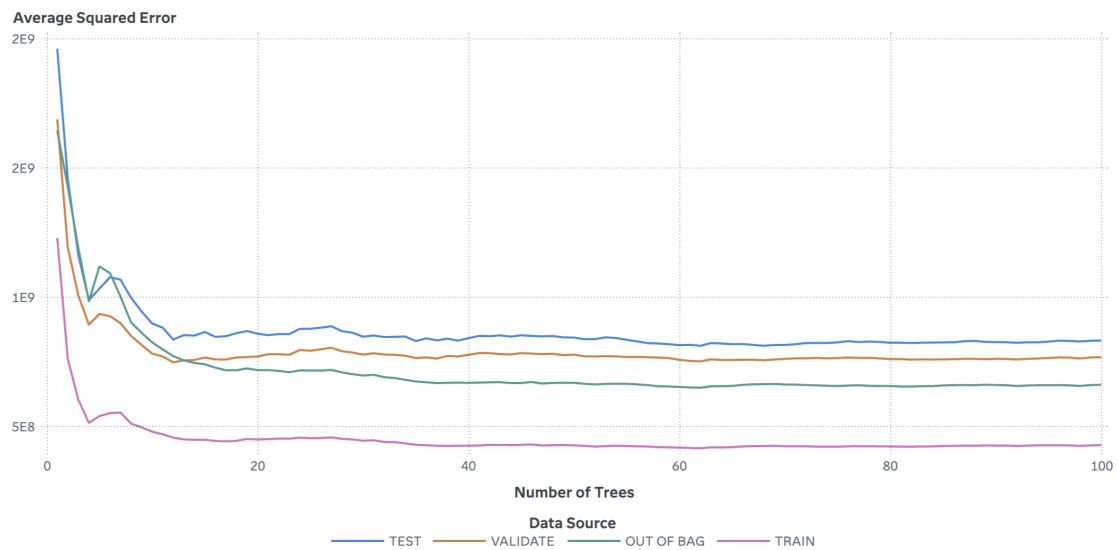


Figure 4.12.: Default Random Forest Error Plot

4.3.2. Tuned Random Forest

In the tuned configuration, the number of trees was increased to 300 and the maximum depth to 25, while the minimum leaf size was increased to 10. The number of variables considered per split remained fixed at three, and the bootstrap sampling rate was reduced to 0.50.

The tuned Random Forest achieved slightly lower predictive performance than the default configuration, with the test R^2 decreasing from 0.971 to 0.960 (Table 4.5). The tuning adjustments increased model bias and reduced model variance, but did not translate into improved predictive performance for this dataset. Instead, the results reflect a typical bias–variance trade-off in ensemble models, where regularisation decreases variance at the expense of reduced fit.

On the test partition, the default Random Forest outperformed the tuned configuration across all error metrics, and this was consistent with its performance on the validation partition. For this dataset, hyperparameter tuning did not improve predictive

Table 4.5.: Fit statistics for the tuned Random Forest model

Partition	N	RMSE (USD)	MAE (USD)	R^2
Train	5,935	27,215	17,120	0.971
Validation	1,979	33,524	20,176	0.957
Test	1,979	33,918	20,444	0.960

performance, suggesting that the default settings already provided an effective balance between bias and variance. The default Random Forest was therefore selected as the champion configuration (Table 4.6).

Table 4.6.: Random Forest model comparison on the test partition

Champion	Configuration	N	RMSE (USD)	R^2
★	Default	1,979	28,869	0.971
	Tuned	1,979	33,918	0.960

4.4. Creating Gradient Boosting Models

Gradient boosting was implemented to assess whether a sequential boosting strategy could improve predictive performance relative to the bagging-based random forest model (Figure 4.13). Unlike random forests, which reduce variance by aggregating independent trees, gradient boosting builds trees sequentially, with each tree correcting the residual errors of the previous ensemble. This iterative structure allows the model to capture subtle nonlinearities and interaction effects, which are particularly relevant in heterogeneous housing markets. Both a default configuration and a tuned configuration were trained within the SAS Viya pipeline, enabling a controlled comparison of boosting performance with and without hyperparameter refinement.

4.4.1. Default Gradient Boosting

A default Gradient Boosting model was trained using the default configuration. Under this setup, the ensemble was constructed from 100 boosting iterations with a learning rate of 0.10 and a subsampling rate of 0.50. The default tree-splitting parameters enforced relatively shallow trees (maximum depth of 4, minimum leaf size of 5) and binary splits, producing weak learners in line with the boosting framework. These settings create a conservative baseline that prioritises bias reduction through sequential residual fitting rather than aggressive variance minimisation.

Model performance was evaluated across the training, validation, and test partitions. The default Gradient Boosting configuration achieved an RMSE of approximately 29.6K dollars and an MAE of roughly 19.0K dollars on the test set, with an R^2 of 0.969 (Table 4.7). Performance was consistent across the validation split (R^2 of around 0.970), indicating stable generalisation. The relatively small gap between the training R^2 (0.980) and the test R^2 suggests that the ensemble captured non-linear price dynamics without

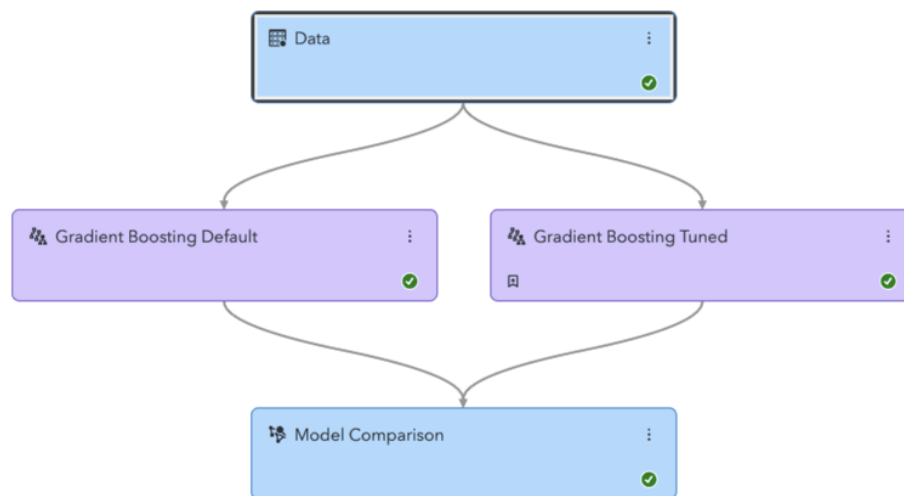


Figure 4.13.: Gradient Boosting Modelling Pipeline

exhibiting severe overfitting. These results confirm that even the default boosting configuration provides strong predictive performance for modelling metropolitan housing prices.

Table 4.7.: Fit statistics for the default Gradient Boosting model

Partition	N	RMSE (USD)	MAE (USD)	R^2
Train	5,935	22,560	16,478	0.980
Validation	1,979	28,326	19,018	0.970
Test	1,979	29,616	19,008	0.969

The error declines sharply during the initial boosting iterations and then gradually stabilises. The training, validation, and test curves remain closely aligned as the ensemble grows (Figure 4.14). This pattern indicates stable generalisation and suggests that the default boosting configuration achieves strong predictive performance without evident overfitting.

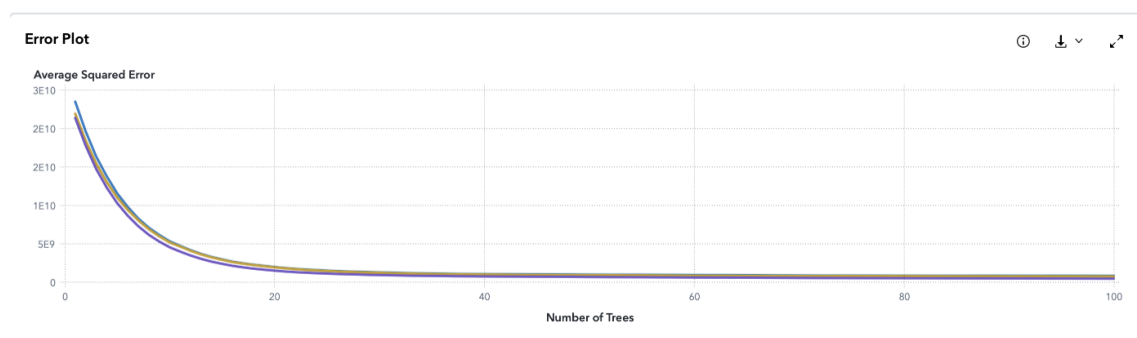


Figure 4.14.: Default Gradient Boosting Error Plot

4.4.2. Tuned Gradient Boosting

In the tuned configuration, the boosting ensemble was adjusted to improve stability and reduce variance. The number of trees was increased to 300, while the learning rate was reduced to 0.05, allowing the model to learn more gradually. A subsampling rate of 0.60 introduced randomness at each iteration, which helps limit overfitting. L2 regularisation was increased to place stronger penalties on complex updates. The tree depth was limited to 6 and the minimum leaf size was increased to 10, which constrains individual trees and reduces sensitivity to noise. The number of interval bins was increased to 200 to allow more detailed splits for continuous predictors. Early stopping was enabled so that training stops when the validation error no longer improves. Overall, the tuning focused on smoother learning behaviour and stronger regularisation compared to the default configuration.

The tuned Gradient Boosting model delivered strong and stable performance across partitions. On the test partition, the RMSE decreased to approximately 19.5K dollars and the MAE to about 12.9K dollars, while the R^2 increased to 0.987 (Table 4.8). The test results were closely aligned with the training (0.995) and validation (0.985) scores, suggesting improved generalisation without signs of overfitting. Overall, the tuned configuration captured more of the non-linear structure in housing prices.

Table 4.8.: Fit statistics for the tuned Gradient Boosting model

Partition	N	RMSE (USD)	MAE (USD)	R^2
Train	5,935	11,946	8,820	0.995
Validation	1,979	19,733	12,959	0.985
Test	1,979	19,461	12,911	0.987

The model comparison confirms that the tuned Gradient Boosting configuration consistently outperforms the default model on the test set, achieving a lower RMSE and a higher R^2 (Table 4.9). This substantial improvement highlights the effectiveness of the tuning choices in increasing predictive accuracy while maintaining stable generalisation.

Table 4.9.: Gradient Boosting model comparison on the test partition

Champion	Configuration	N	RMSE (USD)	R^2
★	Tuned	1,979	19,461	0.987
	Default	1,979	29,616	0.969

4.4.3. HyperSHAP Interpretation of the Tuned Gradient Boosting Model

While the tuned Gradient Boosting model achieved the strongest predictive performance, understanding the factors behind individual predictions remains important. The feature importance measures presented earlier provide a global view of variable influence across the dataset. However, they do not explain how specific variables contribute to individual predictions. To address this limitation, HyperSHAP explanations were generated using the post-training interpretability framework available in SAS Model Studio. HyperSHAP is based on the SHAP framework, which is designed to explain how individual predictors

influence model predictions at the observation level [19]. This method provides local explanations and helps reveal how affordability, rental, and market-related factors influence housing price predictions across different metropolitan areas.

To illustrate these local explanations, three representative observations were selected from the dataset. The HyperSHAP visualisations show whether each variable increases or decreases the predicted housing price and indicate the relative magnitude of its contribution.

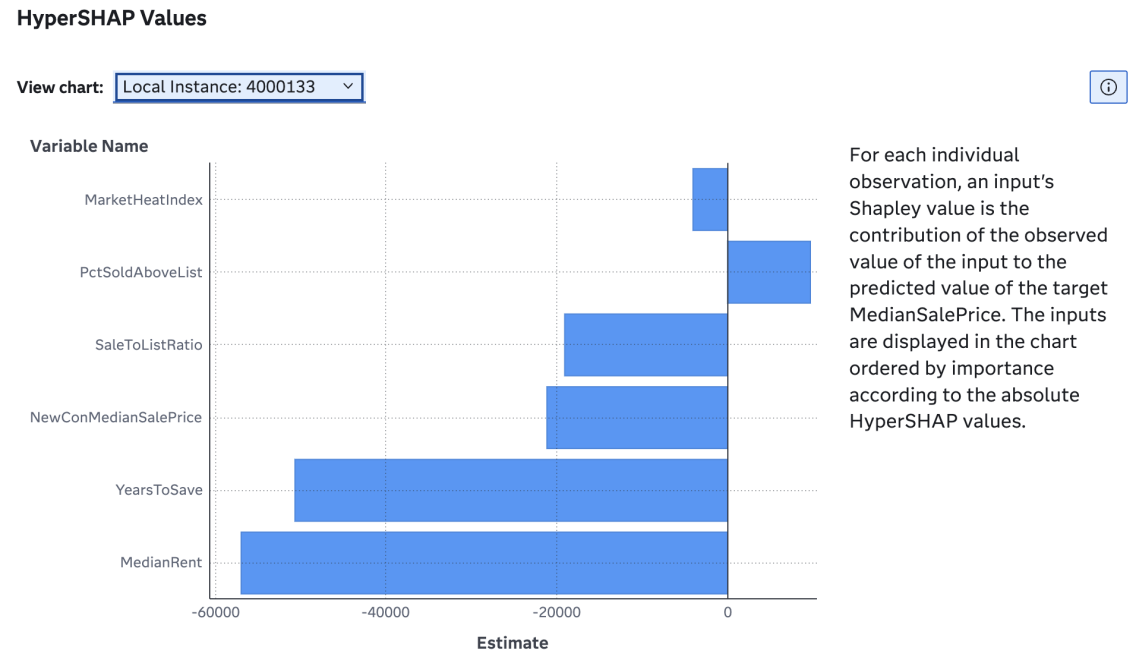


Figure 4.15.: HyperSHAP Explanation for Local Instance 4000133

In this Figure 4.15, affordability-related variables dominate the prediction. **MedianRent** and **YearsToSave** have the largest negative contributions, followed by **NewConMedianSalePrice**. In contrast, **PctSoldAboveList** contributes positively, but its effect is much smaller.

MedianRent generates the largest positive contribution to the prediction Figure 4.16. **NewConMedianSalePrice** and **SaleToListRatio** also contribute positively, while **PctSoldAboveList** has a smaller positive effect. In contrast, **YearsToSave** contributes negatively to the predicted value.

YearsToSave generates the largest negative contribution to the prediction Figure 4.17. **MedianRent** and **NewConMedianSalePrice** also contribute negatively. In contrast, **PctSoldAboveList** provides the strongest positive contribution, while **SaleToListRatio** and **MarketHeatIndex** have relatively small positive effects.

Overall, the HyperSHAP analysis shows that the contribution of individual variables varies across observations. Although affordability-related indicators often have the largest impact on predictions, their magnitude and direction are not identical for every observation. These local explanations complement the global interpretation results and help provide a better understanding of the tuned Gradient Boosting model.

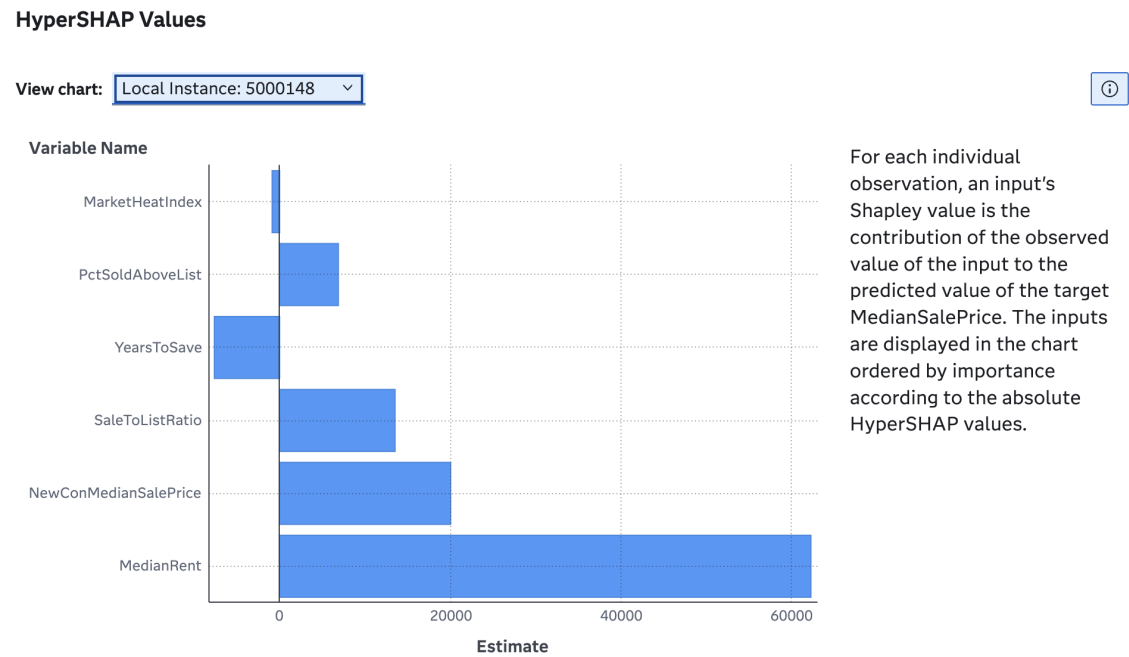


Figure 4.16.: HyperSHAP Explanation for Local Instance 5000148

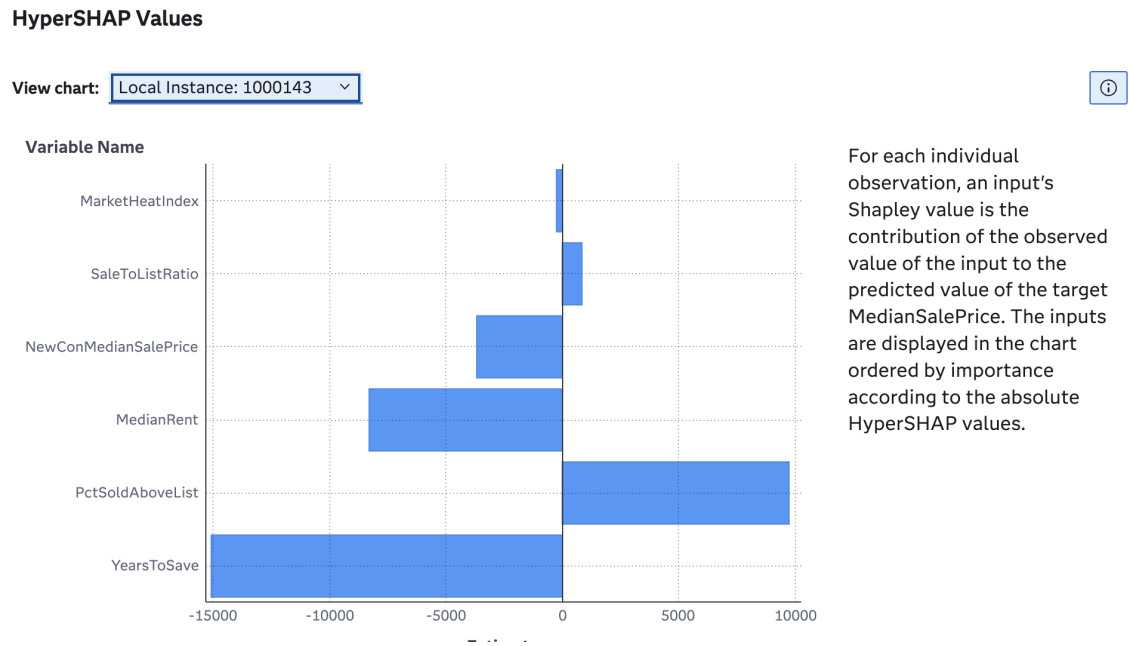


Figure 4.17.: HyperSHAP Explanation for Local Instance 1000143

For the champion model, the tuned Gradient Boosting, the scored output from the SAS Model Studio pipeline was exported to the Explore and Visualize environment. There, a list table was built to inspect the predicted values against the actual sale prices on the test partition. From this output, a set of regions covering the full price range was selected, and these values were used to build Table 4.10. The table reports the actual and predicted median sale price for each region, together with the prediction error in dollars and as a percentage.

Table 4.10.: Actual versus predicted median sale prices for selected test-set observations (tuned Gradient Boosting champion model)

Metropolitan Area	Date	Actual (\$)	Predicted (\$)	Error (\$)	Error (%)
San Jose, CA	Mar 2024	1,560,000	1,486,440	−73,560	−4.7
San Francisco, CA	Nov 2024	1,125,000	1,115,553	−9,447	−0.8
Los Angeles, CA	Dec 2024	935,000	915,954	−19,046	−2.0
Boston, MA	Jun 2024	720,143	653,750	−66,393	−9.2
Seattle, WA	Jun 2023	689,950	637,411	−52,539	−7.6
Portland, OR	Dec 2018	370,000	356,678	−13,322	−3.6
Nashville, TN	Aug 2021	366,950	362,105	−4,845	−1.3
Milwaukee, WI	Jul 2021	260,000	260,354	+354	+0.1
New Orleans, LA	Jun 2021	259,250	257,842	−1,408	−0.5
Omaha, NE	Jan 2024	257,250	258,915	+1,665	+0.6
Montgomery, AL	Aug 2018	129,000	139,439	+10,439	+8.1
Cleveland, OH	Jan 2019	121,975	149,930	+27,955	+22.9

The mid-priced markets give the most accurate predictions. For Milwaukee, New Orleans, and Omaha the error is below one percent. For expensive markets the dollar error is bigger, around \$73K for San Jose and \$66K for Boston, but the percentage error is still small, under ten percent. This is important because the linear model in Section 4.1 made very large errors at high prices, while the Gradient Boosting model stays accurate even above one million dollars.

The biggest errors are at the lowest prices. For cheap markets like Cleveland and Montgomery, the model predicts too high, up to 23% for Cleveland. This happens because tree-based models tend to pull very low values up towards the average. It also fits the HyperSHAP results, where predictions depend mainly on affordability and rent, not on the price level. So two regions with similar prices can get different predictions.

Figure 4.18 presents a scatter plot of the predicted median sale price against the actual sale price for the test partition, generated in the SAS Visual Analytics Explore and Visualize environment using the scored output data. Most points lie very close to the diagonal, indicating strong agreement between predicted and observed values. Although the dispersion increases slightly at higher price levels in absolute terms, the model maintains a strong fit across the full range of housing prices.



Figure 4.18.: Predicted versus actual median sale price on the test partition (tuned Gradient Boosting champion model).

4.5. Model Comparison

The models developed in the previous sections can now be compared directly on the test partition. Table 4.11 summarises the test-set results for all six configurations.

Table 4.11.: Comparison of predictive performance across all evaluated models (test partition)

Family	Configuration	RMSE (USD)	MAE (USD)	R^2
Decision Tree	Default	48,339	32,159	0.918
	Tuned	38,999	25,416	0.947
Random Forest	Default	28,869	17,538	0.971
	Tuned	33,918	20,444	0.960
Gradient Boosting	Default	29,616	19,008	0.969
	Tuned	19,461	12,911	0.987

A clear performance hierarchy emerges from these results. The tuned Gradient Boosting model gave the best results on the test partition and was therefore selected as the champion model. On the test partition it reached an RMSE of about \$19.5K, an MAE of roughly \$12.9K, and an R^2 of 0.987. The default Random Forest follows as the second-best model ($R^2 = 0.971$), just ahead of the default Gradient Boosting configuration ($R^2 = 0.969$). The tuned Random Forest performs slightly worse than its default version, which suggests that the additional regularisation traded variance for bias without any net gain on this dataset.

Both Decision Tree models trail the ensembles. Tuning helps considerably, lowering the RMSE from about \$48.3K to \$39.0K and raising the R^2 from 0.918 to 0.947, but even the tuned tree remains well behind the ensemble methods. This indicates that a single-tree structure struggles to capture the nonlinear relationships and variable interactions that characterise metropolitan housing markets.

Read across families, the results tell a consistent story. Moving from a single tree to a bagging ensemble cuts the test RMSE from roughly \$48.3K to \$28.9K, mainly through variance reduction. Boosting then delivers a second, larger improvement: once properly regularised, the tuned Gradient Boosting model lowers the RMSE by about a further third relative to the best Random Forest. Its advantage therefore comes not only from being an ensemble, but from the sequential way it refines residual errors, which is well suited to markets where affordability, rental pressure, and competitiveness interact in highly nonlinear ways.

Finally, the leading ensembles show little sign of overfitting. For the tuned Gradient Boosting model the training, validation, and test scores stay close together (R^2 of 0.995, 0.985, and 0.987 respectively), pointing to stable generalisation rather than memorisation of the training data. On this basis, the tuned Gradient Boosting model was carried forward as the final model for the interpretation and analysis that follow.

4.6. Chapter summary

This chapter presented the implementation and evaluation of the modelling framework introduced in Chapter 3. Linear regression was used as an interpretable baseline. Three tree-based models were then developed in SAS Viya, using the same 60/20/20 train–validation–test partition: Decision Tree, Random Forest, and Gradient Boosting. The analysis looked at both predictive accuracy and the factors driving housing price variation.

The results showed clear differences between the models. Linear regression and the single decision tree were easy to interpret but less accurate, while the ensemble methods performed much better. Random Forest improved accuracy through bagging, and Gradient Boosting gave the strongest results by capturing more complex nonlinear relationships. The tuned Gradient Boosting model produced the lowest prediction errors and the highest accuracy on the test partition ($R^2 = 0.987$). Across all models, affordability-related variables, especially YearsToSave and MedianRent, were the main drivers of median sale prices.

The next chapter discusses these findings in more detail, answers the research questions from Chapter 1, and outlines directions for future research.

5. Discussion and Further Research

5.1. Conclusion

This study examined urban housing price dynamics across U.S. metropolitan areas using a combination of visual analytics and tree-based machine learning methods. Beyond predictive accuracy, the analysis focused on explaining how affordability, rental pressure, and market conditions interact within heterogeneous housing markets.

A structured modelling workflow was implemented, beginning with an interpretable regression tree to identify key nonlinear patterns and segmentation structures. This was followed by Random Forest and Gradient Boosting models to evaluate the added value of ensemble learning under a unified modelling pipeline with consistent data partitions and evaluation procedures.

The results reveal a clear performance hierarchy across the evaluated models. While the decision tree provided transparent market segmentation, the ensemble methods achieved substantially higher predictive accuracy. Among all evaluated approaches, the tuned Gradient Boosting model delivered the strongest overall performance and the most stable generalisation across data partitions. These findings suggest that sequential boosting is particularly well suited to capturing the complex nonlinear relationships present in housing price dynamics.

Across all models, affordability-related variables were the dominant drivers of housing prices, while short-term market competitiveness indicators played a smaller role. This pattern suggests that long-term affordability constraints matter more than temporary bidding pressure in shaping housing prices.

Overall, the study demonstrates that interpretable ensemble modelling, particularly Gradient Boosting, provides a robust and effective framework for analysing urban housing markets. By combining predictive performance with transparent model interpretation, the approach offers both analytical insight and practical value for understanding the structural drivers of housing price variation. While the analysis was framed within the broader context of SDG 11 (Sustainable Cities and Communities), it remained focused on a concrete and measurable urban domain, illustrating how data-driven methods can inform affordability-related challenges in sustainable city development.

These findings also connect to SDG 11 and its goal of ensuring affordable housing. The results show that long-term affordability plays a larger role in housing prices than short-term market competition. This suggests that affordability indicators can provide a practical way to monitor progress toward sustainable urban housing at the metropolitan level.

5.2. Summary: Answering the Research Questions and Expected Outcomes

This section summarises how the findings of the study address the research questions and expected outcomes presented in Chapter 1.

Research Question 1: What are the primary drivers of housing price variation across U.S. metropolitan regions, and how do affordability and market competitiveness indicators influence housing sale prices?

Affordability-related variables were the strongest factors linked to housing price variation. YearsToSave ranked first across the evaluated models, with MedianRent and NewCon-MedianSalePrice contributing at a secondary level. The market competitiveness indicators, namely PctSoldAboveList, SaleToListRatio, and MarketHeatIndex, added comparatively little. In other words, how long a household needs to save and how high rents are tell us more about sale prices than short-term measures of buyer competition.

Research Question 2: To what extent can tree-based machine learning models outperform traditional linear approaches in predicting urban housing prices?

The tree-based models, and in particular the ensemble methods, outperformed the linear regression baseline. Linear regression provided an interpretable reference point, but its residual diagnostics revealed clear heteroscedasticity and limitations in representing the variance structure of housing prices. The single decision tree contributed mainly interpretability and market segmentation rather than a decisive accuracy gain over the baseline, whereas the Random Forest and Gradient Boosting ensembles were substantially more accurate. The tuned Gradient Boosting model gave the strongest result, reducing the test RMSE from approximately \$48.3K obtained by the default Decision Tree to roughly \$19.5K and reaching an R^2 of 0.987. These results indicate that tree-based ensemble approaches are better suited for modelling the nonlinear relationships present in metropolitan housing market data.

Research Question 3: How do housing prices and affordability metrics evolve temporally across U.S. regions between 2018 and 2025?

Since the tree-based models treat observations independently and do not model temporal dynamics, this research question is addressed through the exploratory temporal analysis presented earlier in the study. The temporal analysis revealed a clear increase in both housing prices and rental values throughout the study period. The strongest growth in sale prices was observed between 2021 and 2022, with rents rising particularly sharply from late 2021 onwards, after which both remained at relatively high levels. The analysis also highlighted notable regional differences, with coastal metropolitan areas generally exhibiting higher housing prices and stronger affordability pressures than many non-coastal regions. These findings reflect the uneven nature of housing market conditions across U.S. metropolitan regions.

Research Question 4: What interpretable and meaningful patterns can be extracted from tree-based models through visual analytics and rule-based interpretation?

The visual analytics and model interpretation outputs provided a clearer understanding of housing market segmentation. Regression trees, variable importance measures,

node rules, icicle visualisations, and HyperSHAP explanations consistently showed that affordability-related variables play the central role in housing market segmentation. HyperSHAP further revealed that the magnitude and direction of variable contributions can vary considerably across individual metropolitan observations.

Expected Outcomes

Overall, the expected outcomes defined in Chapter 1 were achieved. The study developed a data-driven and interpretable analytical framework for analysing urban housing markets, identified the main drivers of housing price variation, demonstrated the advantages of tree-based machine learning methods over traditional linear approaches, and provided interpretable insights into housing market segmentation through visual analytics. The results also confirmed the value of ensemble methods, particularly Gradient Boosting, for modelling housing price variation while maintaining a meaningful level of interpretability.

5.3. Limitations

While this study provides useful and interpretable insights into housing price dynamics, a few limitations should be acknowledged when interpreting the results.

The first limitation concerns the data. All variables come from a single source, Zillow Research, and the study only covers U.S. metropolitan areas. This keeps the data consistent, but it also means the results depend on how Zillow measures these indicators. As a result, the findings may not apply to rural or smaller housing markets, which often behave differently. Future studies could validate these findings using additional data sources and a broader range of geographic regions.

The second limitation concerns the variables used. The models are based on affordability, rental, and competitiveness indicators. However, other factors that affect housing prices, such as interest rates, household income, and housing supply, were not included. These factors vary substantially across regions and may influence housing prices beyond the affordability, rental, and competitiveness indicators examined in this study. As a result, some determinants of housing prices are not fully captured by the current modelling framework. Adding such indicators would be a useful extension.

The third limitation concerns interpretation. The models identify which variables are most strongly linked to housing prices, but this reflects association, not causation. In other words, the analysis shows which factors are related to price changes, but it does not establish that they cause them.

Finally, the models are static. The tree-based models treat each observation independently and do not use the time structure of the data. This means they explain how prices relate to market indicators, but not how prices change over time. Future work could explore time-series methods to capture how housing markets evolve over time.

In addition, all results are based on a single train-validation-test partition. Future research could evaluate model stability through repeated resampling or cross-validation procedures.

5.4. Future Research Directions

The limitations discussed above also point to several directions for future research.

First, the analysis relies on Zillow data for U.S. metropolitan areas. Future research could apply the same framework to other regions or international cities. Testing the models in areas with different economic conditions would help assess the generalisability of the framework across different housing markets.

Second, the models could include more explanatory variables. Adding factors such as interest rates, household income, and housing supply could explain more of the variation in prices. Future studies could also extend the sustainability perspective of the framework by incorporating indicators related to environmental quality, transport accessibility, or social inequality. Such additions could provide a more comprehensive view of long-term urban development within the context of SDG 11 (Sustainable Cities and Communities).

Third, future work could go beyond the static modelling approach used here. Time-series methods could be applied to capture how housing prices change over time and to model temporal patterns more explicitly.

Finally, the interpretability of the models could be improved further. HyperSHAP was used to explain individual predictions in this study. Future research could complement this analysis with techniques such as partial dependence plots and accumulated local effects. These methods may provide additional insight into nonlinear relationships and interaction effects within housing markets.

Appendices

A. Source Files

```
1 import pandas as pd
2 import numpy as np
3 import seaborn as sns
4 import matplotlib.pyplot as plt
5 import scipy.stats as stats
6 from scipy.stats import pearsonr
7 from matplotlib.colors import LinearSegmentedColormap
8
9 median = pd.read_csv("Median_sale_price.csv")
10 pct_above = pd.read_csv("Percent_of_homes_sold_above_list.csv")
11 rent = pd.read_csv("Zori_median_rent.csv")
12 market_heat = pd.read_csv("Market_heat.csv")
13 affordability = pd.read_csv("Affordability_years_to_save.csv")
14 sale_list = pd.read_csv("Mean_sale_to_list_ratio.csv")
15 new_cons_sale_price = pd.read_csv("New_con_median_sale_price.csv")
16
17 # Inspect the structure of each dataset before merging to check they have the
18   same columns
19 raw_datasets = {
20     "Median Sale Price": median,
21     "Pct Sold Above List": pct_above,
22     "Median Rent": rent,
23     "Market Heat Index": market_heat,
24     "Years to Save": affordability,
25     "Sale-to-List Ratio": sale_list,
26     "New Construction Sale Price": new_cons_sale_price,
27 }
28
29 for name, df in raw_datasets.items():
30     print(f"\n{name} -> shape: {df.shape}")
31     print("First columns:", df.columns[:6].tolist())
32     print(df.head(3))
33
34 # Define a melt function
35 def melt_df(df, value_name):
36     date_columns = df.columns[5:]
37     melted = df.melt(
38         id_vars=["RegionID", "RegionName", "RegionType", "StateName"],
39         value_vars=date_columns,
40         var_name="Date",
41         value_name=value_name
42     )
43     melted["Date"] = pd.to_datetime(melted["Date"])
44     return melted
45
46 median_melted = melt_df(median, "MedianSalePrice")
47 pct_melted = melt_df(pct_above, "PctSoldAboveList")
48 rent_melted = melt_df(rent, "MedianRent")
49 market_heat_melted = melt_df(market_heat, "MarketHeatIndex")
50 affordability_melted = melt_df(affordability, "YearsToSave")
51 sale_list_melted = melt_df(sale_list, "SaleToListRatio")
```

```

51 new_cons_sale_price_melted = melt_df(new_cons_sale_price, "
    NewConMedianSalePrice")
52
53 merged = median_melted.copy()
54 merged = merged.merge(pct_melted, on=["RegionID", "RegionName", "RegionType", "
    StateName", "Date"])
55 merged = merged.merge(rent_melted, on=["RegionID", "RegionName", "RegionType",
    "StateName", "Date"])
56 merged = merged.merge(market_heat_melted, on=["RegionID", "RegionName", "
    RegionType", "StateName", "Date"])
57 merged = merged.merge(affordability_melted, on=["RegionID", "RegionName", "
    RegionType", "StateName", "Date"])
58 merged = merged.merge(sale_list_melted, on=["RegionID", "RegionName", "
    RegionType", "StateName", "Date"])
59 merged = merged.merge(new_cons_sale_price_melted, on=["RegionID", "RegionName",
    "RegionType", "StateName", "Date"])
60
61 print(merged.head())
62
63 print("Shape:", merged.shape)
64
65 print("\nColumn Types:\n", merged.dtypes)
66
67 missing_values = merged.isnull().sum()
68 missing_values = missing_values[missing_values > 0].sort_values(ascending=False
    )
69 print("\nMissing values per column:\n", missing_values)
70
71 plt.figure(figsize=(12, 6))
72 sns.heatmap(merged.isnull(), cbar=False, cmap="viridis")
73 plt.title("Missing Values Heatmap")
74 plt.show()
75
76 # Drop rows with missing values (not NewConMedianSalePrice)
77 core_columns = ["PctSoldAboveList", "SaleToListRatio", "MedianRent", "StateName
    "]
78 merged_cleaned = merged.dropna(subset=core_columns)
79
80 print("\nRows before drop:", merged.shape[0])
81 print("Rows after drop (excluding only key variables):", merged_cleaned.shape
    [0])
82
83 print("\nDuplicate Rows:", merged.duplicated().sum())
84
85 numeric_cols = merged_cleaned.select_dtypes(include=['float64', 'int64']).drop(
    columns=["RegionID"])
86
87 for col in numeric_cols.columns:
88     plt.figure(figsize=(8, 4))
89     sns.boxplot(x=merged_cleaned[col])
90     plt.title(f' Box Plot for {col}')
91     plt.xlabel(col)
92     plt.tight_layout()
93     plt.show()
94
95 # Descriptive Statistics
96 desc_stats = merged_cleaned.describe()
97 print(desc_stats)
98
99 # Histogram for numeric columns
100 numeric_cols = merged_cleaned.select_dtypes(include=['float64', 'int64']).drop(

```

```

    columns=["RegionID"])
101
102 numeric_cols.hist(bins=50, figsize=(20, 15), edgecolor='black')
103 plt.tight_layout()
104 plt.show()
105
106 # Plot Q-Q plots
107 for col in numeric_cols.columns:
108     plt.figure(figsize=(6, 4))
109     stats.probplot(numeric_cols[col].dropna(), dist="norm", plot=plt)
110     plt.title(f'Q-Q Plot for {col}')
111     plt.grid(True)
112     plt.tight_layout()
113     plt.show()
114
115 # Correlation matrix
116 cols = [
117     "MarketHeatIndex",
118     "MedianRent",
119     "MedianSalePrice",
120     "NewConMedianSalePrice",
121     "PctSoldAboveList",
122     "SaleToListRatio",
123     "YearsToSave",
124 ]
125
126 corr = merged_cleaned[cols].corr()
127
128 blue_scale = LinearSegmentedColormap.from_list("blue_scale", ["#d9e9f7", "#4
    da3f5"])
129
130 mask = np.triu(np.ones_like(corr, dtype=bool))
131 corr_plot = corr.copy()
132 corr_plot[mask] = np.nan
133
134 fig, ax = plt.subplots(figsize=(10, 8))
135 im = ax.imshow(corr_plot, cmap=blue_scale, vmin=0, vmax=1)
136
137 ax.set_xticks(np.arange(len(cols)))
138 ax.set_yticks(np.arange(len(cols)))
139 ax.set_xticklabels(cols, rotation=45, ha="right")
140 ax.set_yticklabels(cols)
141
142 for i in range(len(cols)):
143     for j in range(len(cols)):
144         if not np.isnan(corr_plot.iloc[i, j]):
145             value = corr_plot.iloc[i, j]
146
147             text_color = "white" if value > 0.65 else "black"
148             ax.text(j, i, f"{value:.2f}", ha="center", va="center",
149                     fontsize=10, fontweight="bold", color=text_color)
150
151 ax.set_xticks(np.arange(-.5, len(cols), 1), minor=True)
152 ax.set_yticks(np.arange(-.5, len(cols), 1), minor=True)
153 ax.grid(which="minor", color="white", linestyle="-", linewidth=2)
154 ax.tick_params(which="minor", bottom=False, left=False)
155
156
157 cbar = plt.colorbar(im)
158 cbar.set_label("Correlation Coefficient")
159

```

```

160 plt.title("Correlation Matrix of Housing Market Variables", fontsize=15, pad
    =20)
161
162 plt.tight_layout()
163
164 plt.savefig("correlation_matrix_thesis.png", dpi=300, bbox_inches="tight")
165 plt.show()
166
167 from scipy.stats import pearsonr
168
169 # Pearson correlation tests for the key relationships
170 subset_corr = merged_cleaned[['MedianSalePrice', 'NewConMedianSalePrice']].
    dropna()
171 r1, p1 = pearsonr(subset_corr['MedianSalePrice'], subset_corr['
    NewConMedianSalePrice'])
172 print("Correlation between Median Sale Price and NewConMedianSalePrice:")
173 print(f"Pearson r = {r1:.2f} | p-value = {p1:.4f}")
174
175 subset_corr = merged_cleaned[['MedianSalePrice', 'SaleToListRatio']].dropna()
176 r1, p1 = pearsonr(subset_corr['MedianSalePrice'], subset_corr['SaleToListRatio'
    ])
177 print("\nCorrelation between Median Sale Price and SaleToListRatio:")
178 print(f"Pearson r = {r1:.2f} | p-value = {p1:.4f}")
179
180 subset_corr = merged_cleaned[['MarketHeatIndex', 'PctSoldAboveList']].dropna()
181 r1, p1 = pearsonr(subset_corr['MarketHeatIndex'], subset_corr['PctSoldAboveList
    '])
182 print("\nCorrelation between MarketHeatIndex and PctSoldAboveList:")
183 print(f"Pearson r = {r1:.2f} | p-value = {p1:.4f}")
184
185 # Time Series Plot: Average median sale price over time
186 merged_cleaned["Date"] = pd.to_datetime(merged_cleaned["Date"])
187
188 price_trend = merged_cleaned.groupby("Date")["MedianSalePrice"].mean()
189
190 plt.figure(figsize=(12, 6))
191 plt.plot(price_trend.index, price_trend.values, marker='o', label="Median Sale
    Price", color='royalblue')
192 plt.title("Average Median Sale Price Over Time in the U.S.")
193 plt.xlabel("Date")
194 plt.ylabel("Median Sale Price ($)")
195 plt.grid(True)
196 plt.legend()
197 plt.tight_layout()
198 plt.show()
199
200 # Average median rent over time
201 rent_trend = merged_cleaned.groupby("Date")["MedianRent"].mean()
202
203 plt.figure(figsize=(12, 6))
204 plt.plot(rent_trend.index, rent_trend.values, marker='o', label="Median Rent",
    color='seagreen')
205 plt.title("Average Median Rent Over Time in the U.S.")
206 plt.xlabel("Date")
207 plt.ylabel("Median Rent ($)")
208 plt.grid(True)
209 plt.legend()
210 plt.tight_layout()
211 plt.show()
212
213 # Average sale price and rent by state, sorted by median sale price

```

```

214 state_group = merged_cleaned.groupby("StateName")[["MedianSalePrice", "
    MedianRent"]].mean()
215
216 state_group = state_group.sort_values("MedianSalePrice", ascending=False)
217
218 plt.figure(figsize=(18, 6))
219 bar_width = 0.4
220 states = state_group.index
221 x = range(len(states))
222
223 plt.bar([i for i in x], state_group["MedianRent"], width=bar_width, label="
    Median Rent", color="mediumseagreen", zorder=2)
224
225 plt.bar([i + bar_width for i in x], state_group["MedianSalePrice"], width=
    bar_width, label="Median Sale Price", color="steelblue", zorder=1)
226
227 plt.xlabel("State")
228 plt.ylabel("Amount ($)")
229 plt.title("Average Median Sale Price and Median Rent by State")
230 plt.xticks([i + bar_width / 2 for i in x], states, rotation=45, ha='right')
231 plt.legend()
232
233 plt.grid(axis='y', linestyle='--', alpha=0.5, zorder=0)
234 plt.tight_layout()
235 plt.show()
236
237 # Average sale price and rent by state, sorted by median rent
238 state_group = merged_cleaned.groupby("StateName")[["MedianSalePrice", "
    MedianRent"]].mean()
239
240 state_group = state_group.sort_values("MedianRent", ascending=False)
241
242 plt.figure(figsize=(18, 6))
243 bar_width = 0.4
244 states = state_group.index
245 x = range(len(states))
246
247 plt.bar([i for i in x], state_group["MedianRent"], width=bar_width, label="
    Median Rent", color="mediumseagreen", zorder=2)
248
249 plt.bar([i + bar_width for i in x], state_group["MedianSalePrice"], width=
    bar_width, label="Median Sale Price", color="steelblue", zorder=1)
250
251 plt.xlabel("State")
252 plt.ylabel("Amount ($)")
253 plt.title("Average Median Sale Price and Median Rent by State")
254 plt.xticks([i + bar_width / 2 for i in x], states, rotation=45, ha='right')
255 plt.legend()
256
257 plt.grid(axis='y', linestyle='--', alpha=0.5, zorder=0)
258 plt.tight_layout()
259 plt.show()
260
261 merged_cleaned.to_csv("merged_cleaned.csv", index=False)

```

Listing A.1: Data Preparation and Exploratory Data Analysis

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